Learning to relate images

Roland Memisevic

August 3, 2011

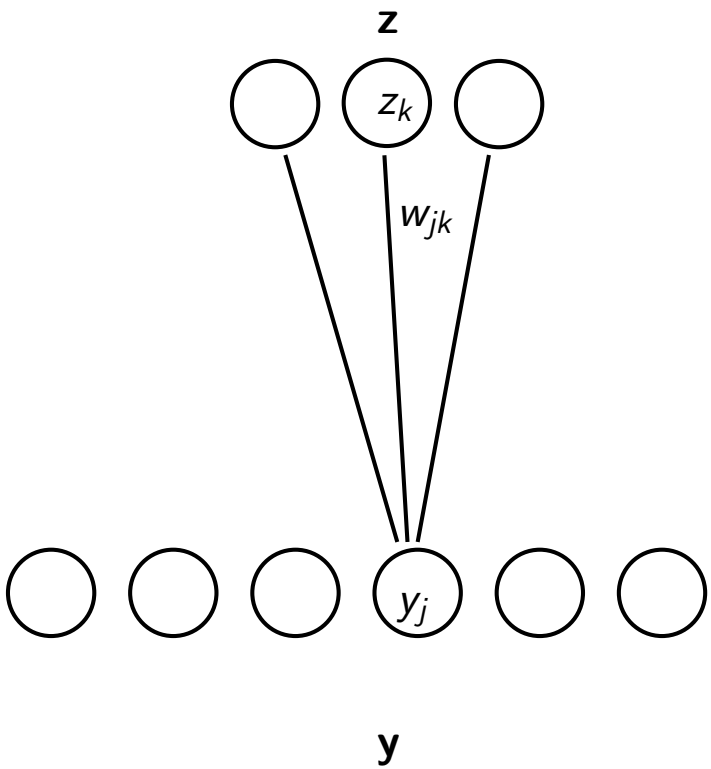
Images are not random

- The set of all natural images ("all photos ever taken") is much smaller than the set of *all possible* images.
- Pixels don't vary randomly across an image.
- *Learning about natural image statistics* is therefore a common task in vision, biological and artificial.
- Common applications are recognition, compression, in-painting, de-noising, etc.

Learning image statistics

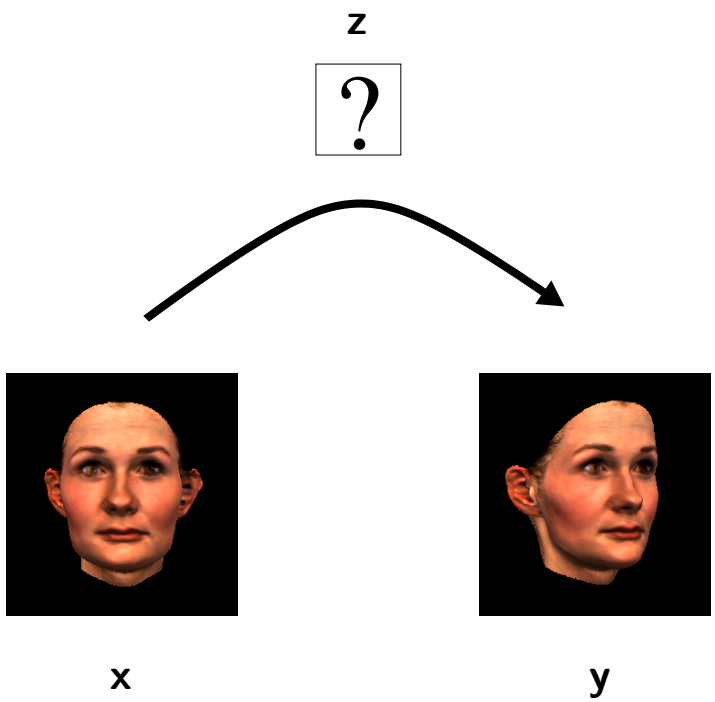
"Dictionary Learning", "Receptive Field Learning", "Sparse Coding", "Feature Learning"

Feature Learning and bi-partite models



- By far the most common modeling framework: Let data and hidden variables populate a **bi-partite network**.
- Examples: PCA, ICA, Sparse Coding, auto-encoders, Restricted Boltzmann Machines, K-means, etc. etc. etc.

Learning how images *change*



- Here: Learn to encode how images **change** (for example, in videos).
- Motivation: *Correspondence* is one of the most ubiquitous problems in vision.

A model of transformations

- For *binary* $\mathbf{x}, \mathbf{y}, \mathbf{z}$:
- Define the *energy-function*:

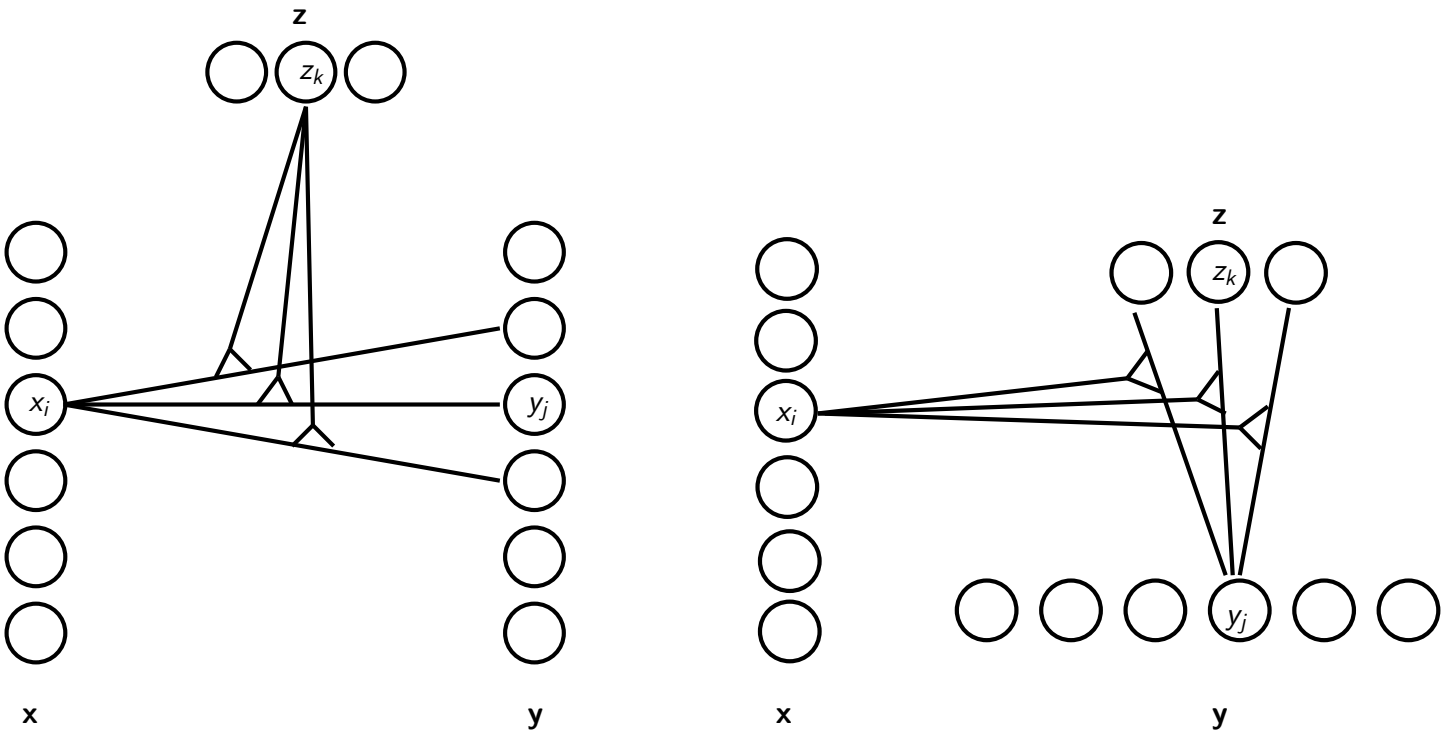
$$E(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \sum_{ijk} w_{ijk} x_i y_j z_k$$

- Get probabilities by exponentiating and normalizing:

$$\begin{aligned} p(\mathbf{y}, \mathbf{z}|\mathbf{x}) &= \frac{1}{\phi(\mathbf{x})} \exp(E(\mathbf{x}, \mathbf{y}, \mathbf{z})) \\ \phi(\mathbf{x}) &= \sum_{\mathbf{y},\mathbf{z}} E(\mathbf{x}, \mathbf{y}, \mathbf{z}) \end{aligned}$$

- From this we get $p(\mathbf{y}|\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{y}, \mathbf{z}|\mathbf{x}) \quad (\propto \prod_k f_k(\mathbf{y}))$
(Normalization not tractable, but sampling easy.)

Three-way interactions



1. Combinatorial Mixture of Experts.
2. Input-modulated filters.

Inference and Learning

Inference

- Inference is simple and efficient, because of statistical independencies:

$$p(z_k = 1|\mathbf{y}, \mathbf{x}) = \frac{1}{1 + \exp(-\sum_{ij} x_i y_j w_{ijk})}$$

$$p(y_j = 1|\mathbf{z}, \mathbf{x}) = \frac{1}{1 + \exp(-\sum_{ij} x_i h_k w_{ijk})}$$

Learning

- Maximize $\sum_\alpha \log p(\mathbf{y}^\alpha|\mathbf{x}^\alpha)$
- Gibbs sampling. Efficient because of statistical independencies.
- "Contrastive divergence" $\rightarrow$ Statistical Hebbian learning rule.

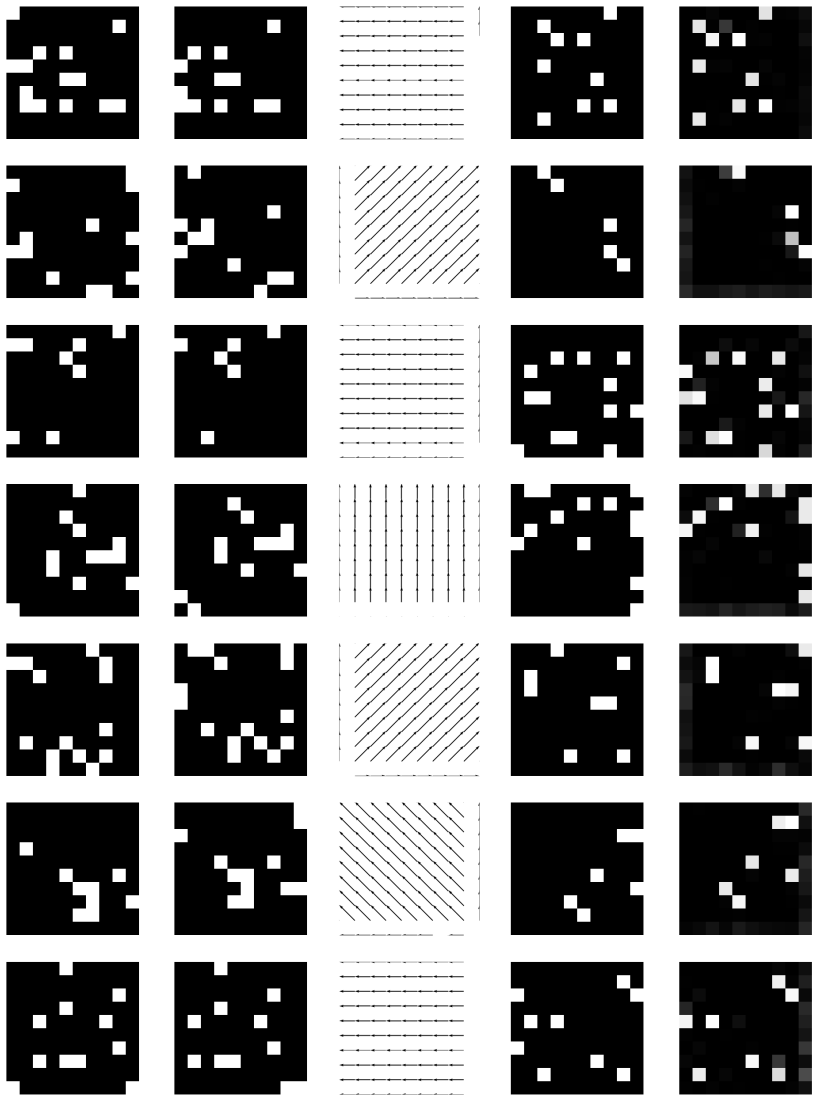
Technicalities

- In practice, add *bias terms* $\sum_k h_k w_k^{\mathrm{h}}$, $\sum_j y_j w_j^{\mathrm{y}}$.
- We can extend the model, so that conditionals are from the Exponential Family.
- This amounts to adding other types of "bias term".
- Can also model the *joint* $p(\mathbf{x}, \mathbf{y})$ instead of the conditional.
- Other, non-probabilistic tri-partite models possible.
- For large images, *convolutional* (Roth, Black; 2005).

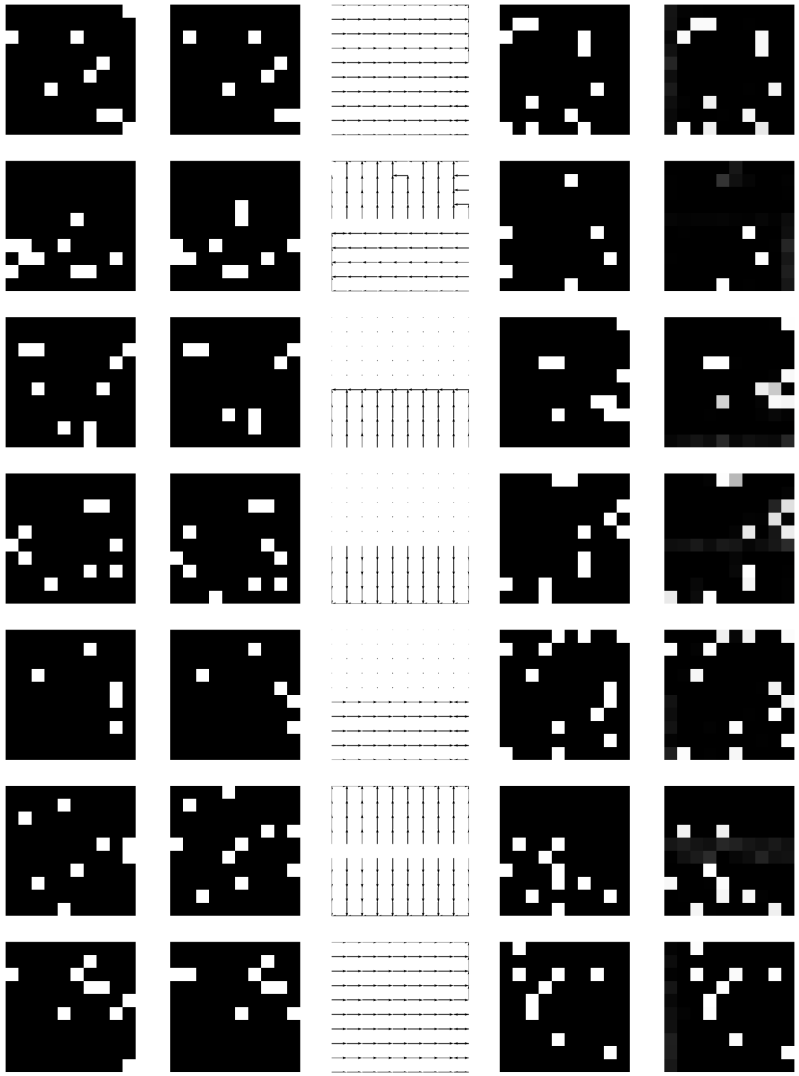
Related models

- Bilinear models (Tenenbaum, Freeman), (Olshausen, et al.)
- "Mapping units", Hinton, 1981
- "Dynamic mappings", v.d. Malsburg, 1981
- Two-level ICA, ISA, (Hyvärinen, et al.)
- Learning quadrature pairs (Bethge, et al., 2007)

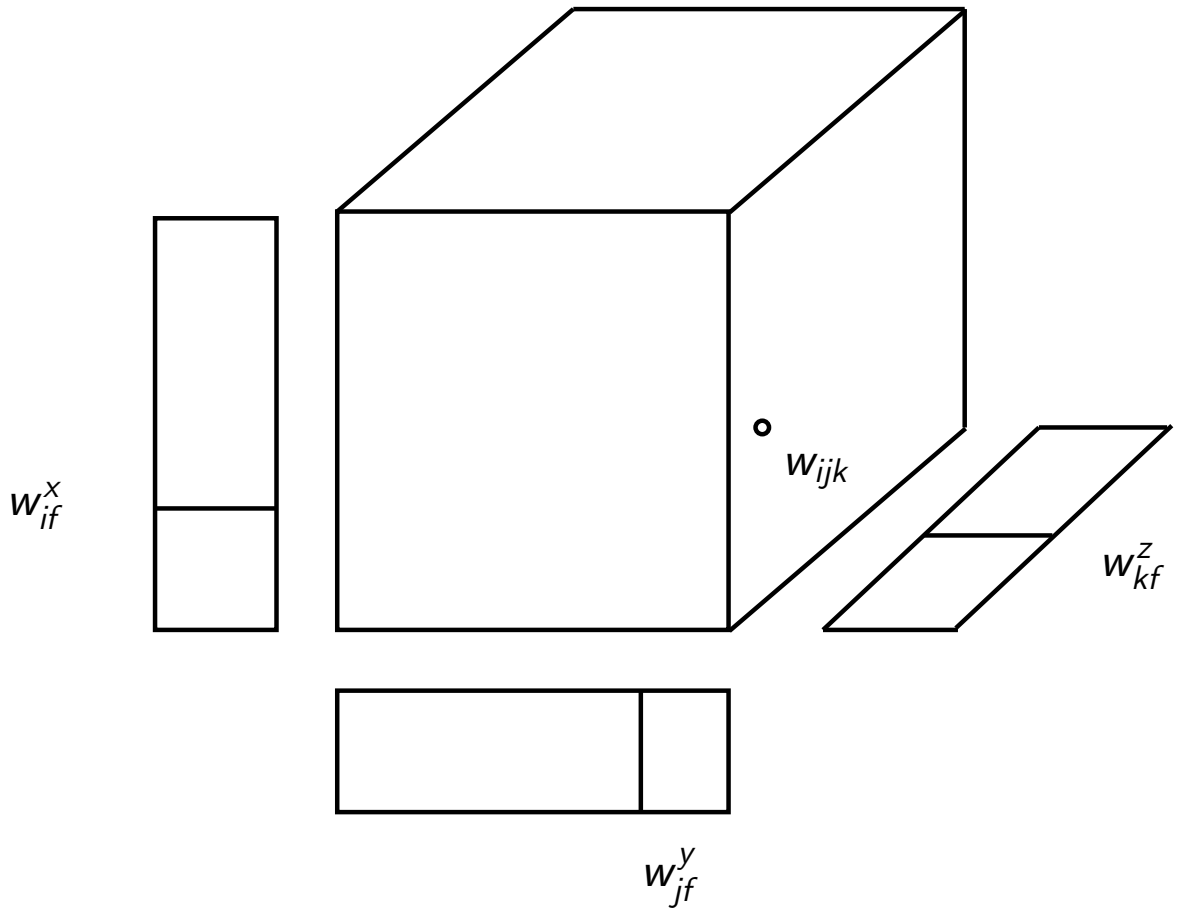
Toy example: "Optical flow"



“Combinatorial flow”

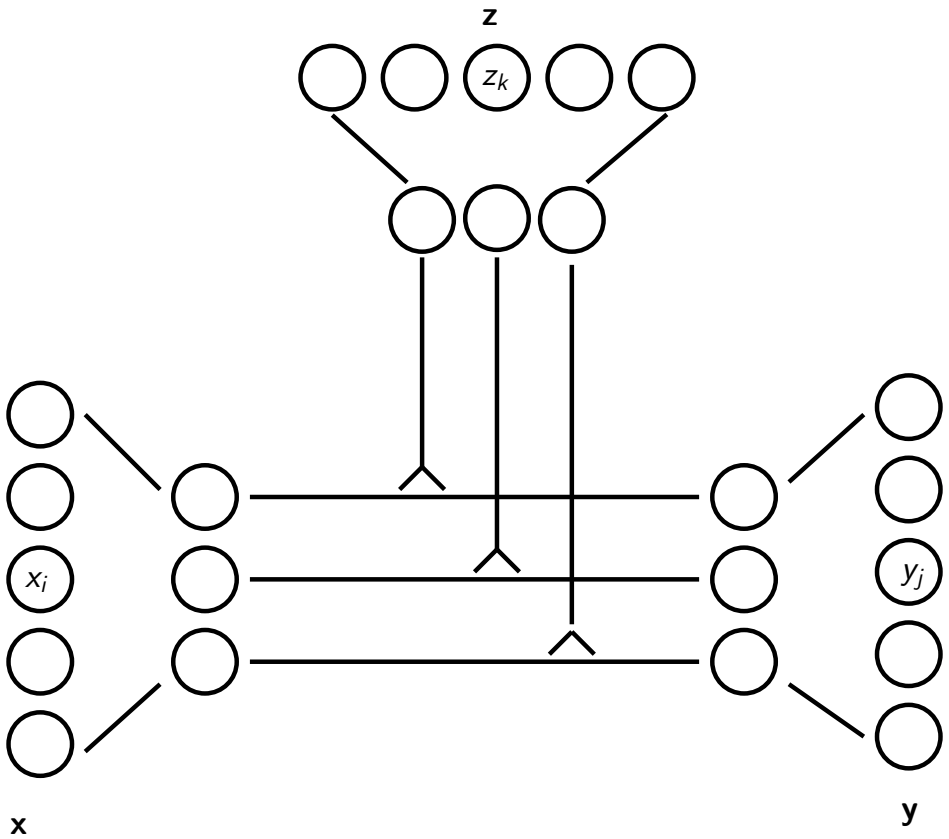


Factorization



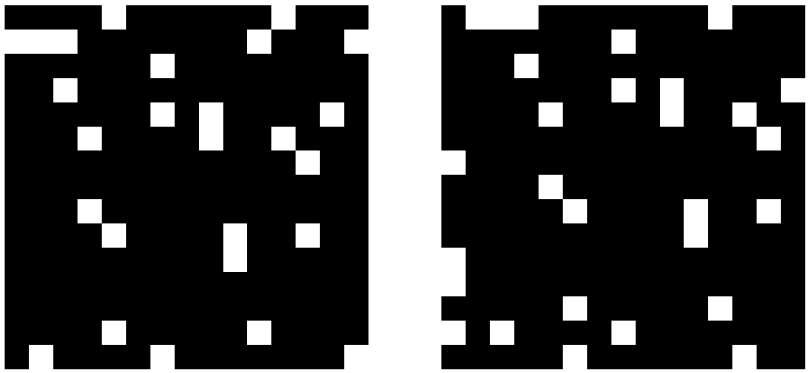
$$E = \sum_{ijk} w_{ijk} x_i y_j z_k = \sum_{ijk} \left(\sum_f w^x_{if} w^y_{jf} w^z_{kf} \right) x_i y_j z_k$$

Factorization is *filter matching*



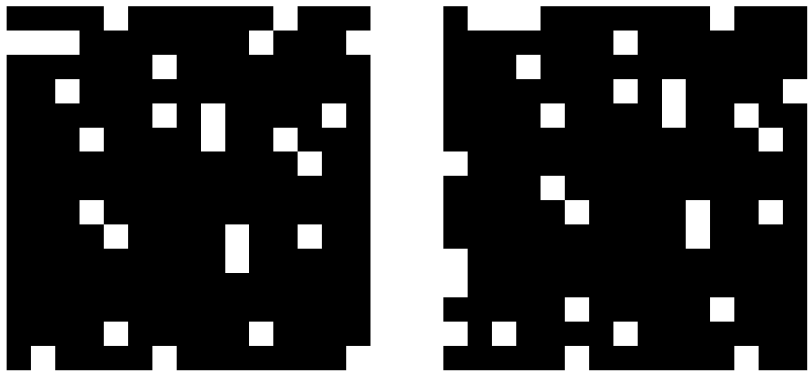
$$E = \sum_{ijk}(\sum_{f} w^x_{if} w^y_{jf} w^z_{kf}) x_i y_j z_k = \sum_{f}(\sum_{i} w^x_{if} x_i)(\sum_{j} w^y_{jf} y_j)(\sum_{k} w^z_{kf} z_k)$$

Learning relational codes



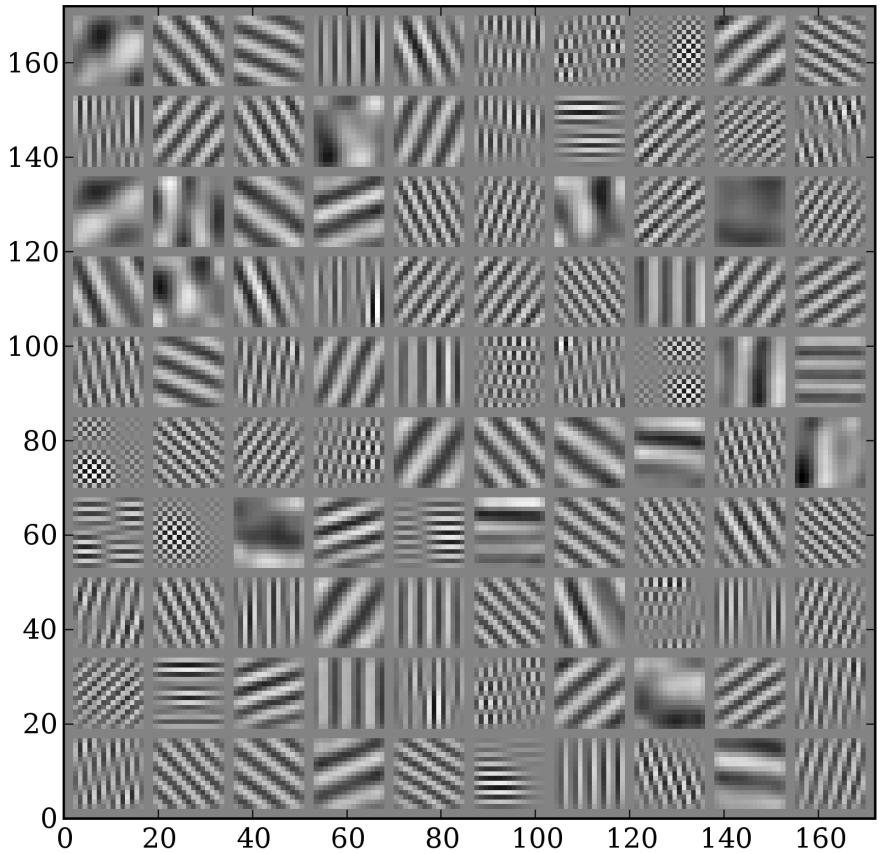
- There is no structure in these images.

Learning relational codes

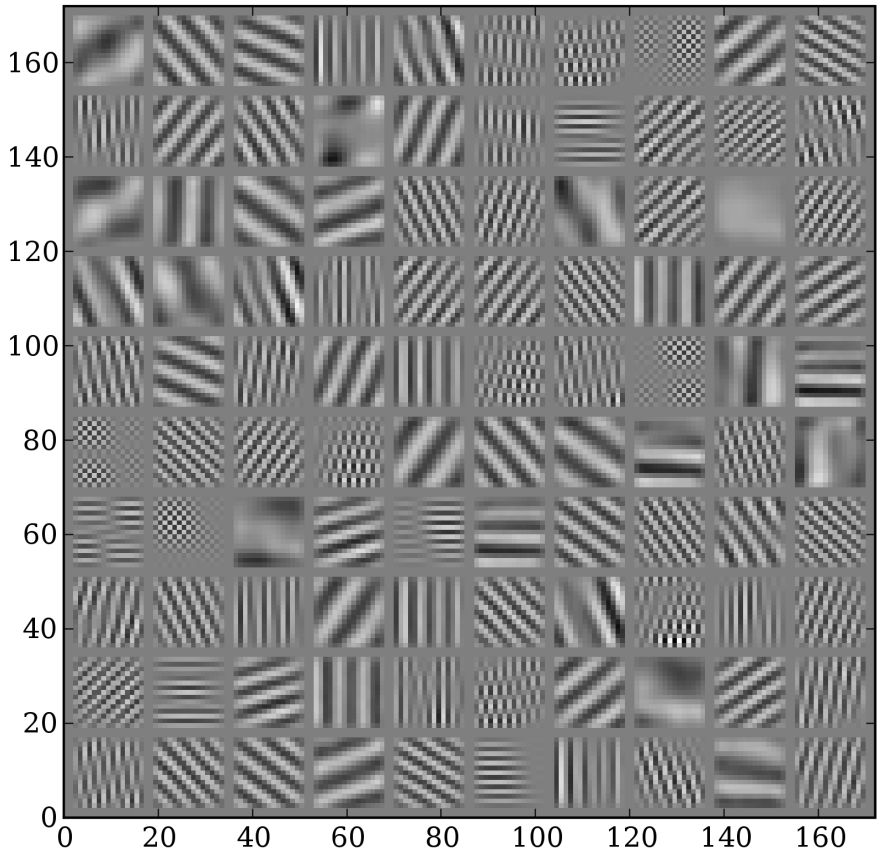


- There is no structure in these images.
- But in *how they change*.

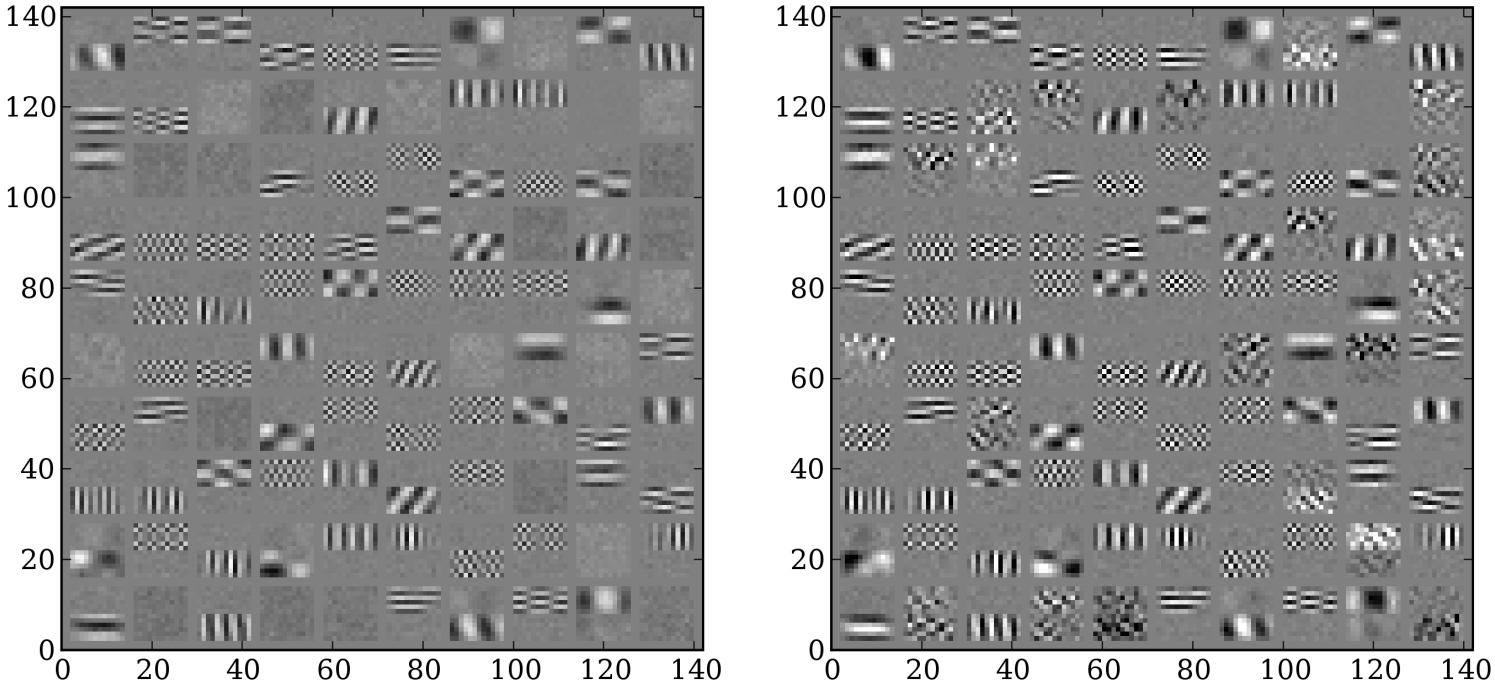
Learned filters w_{if}^{x}



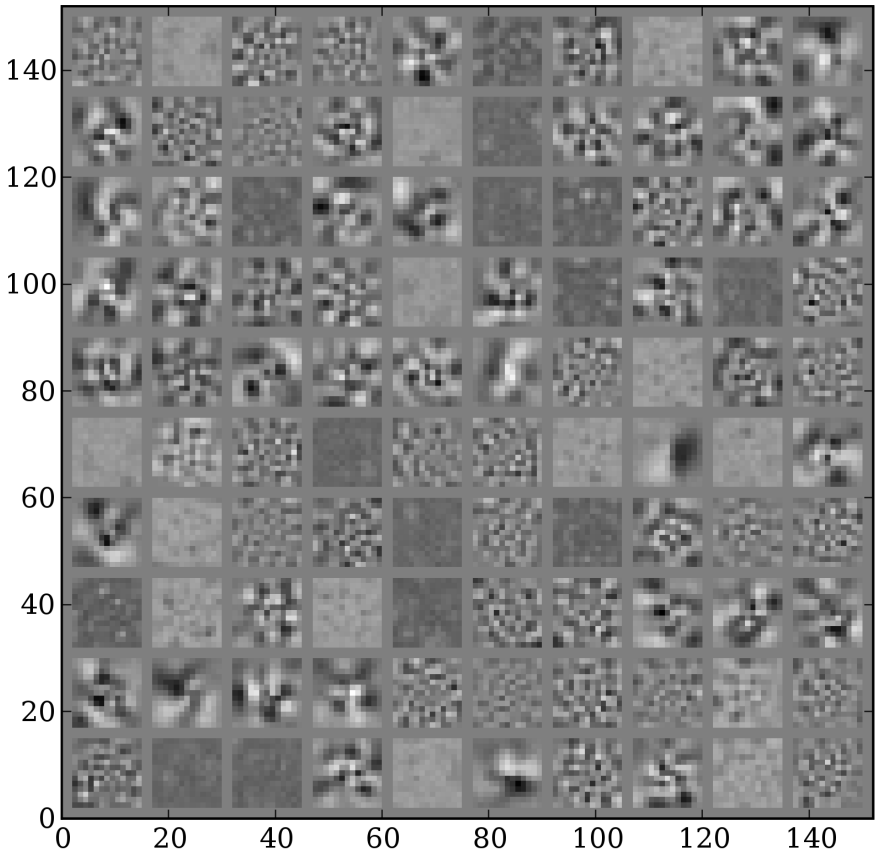
Learned filters w^y_{jf}



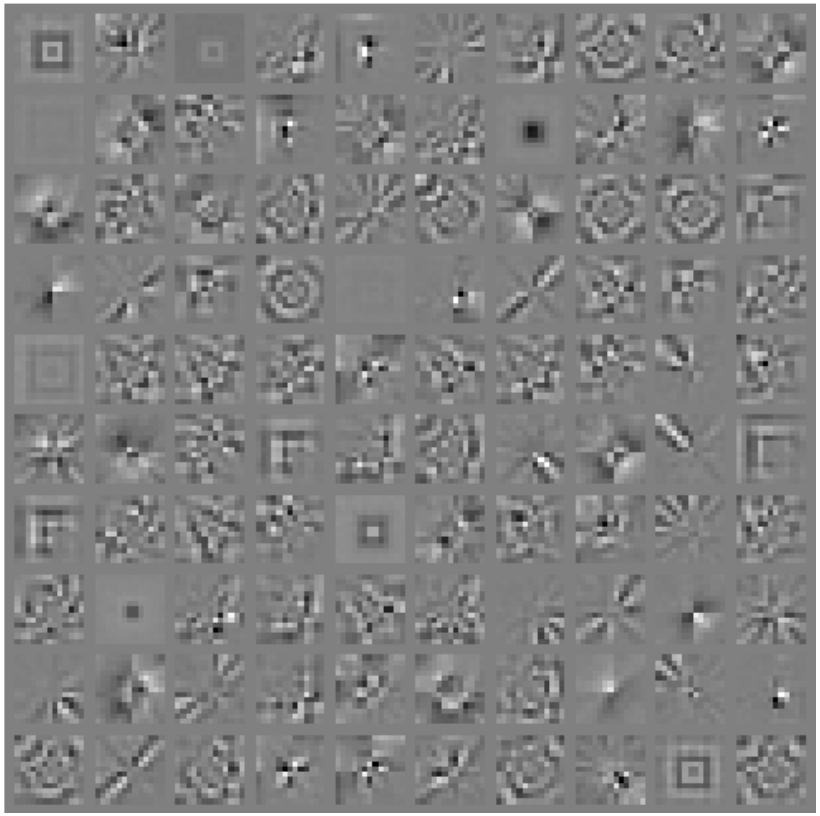
Filters learned from split-screen shifts



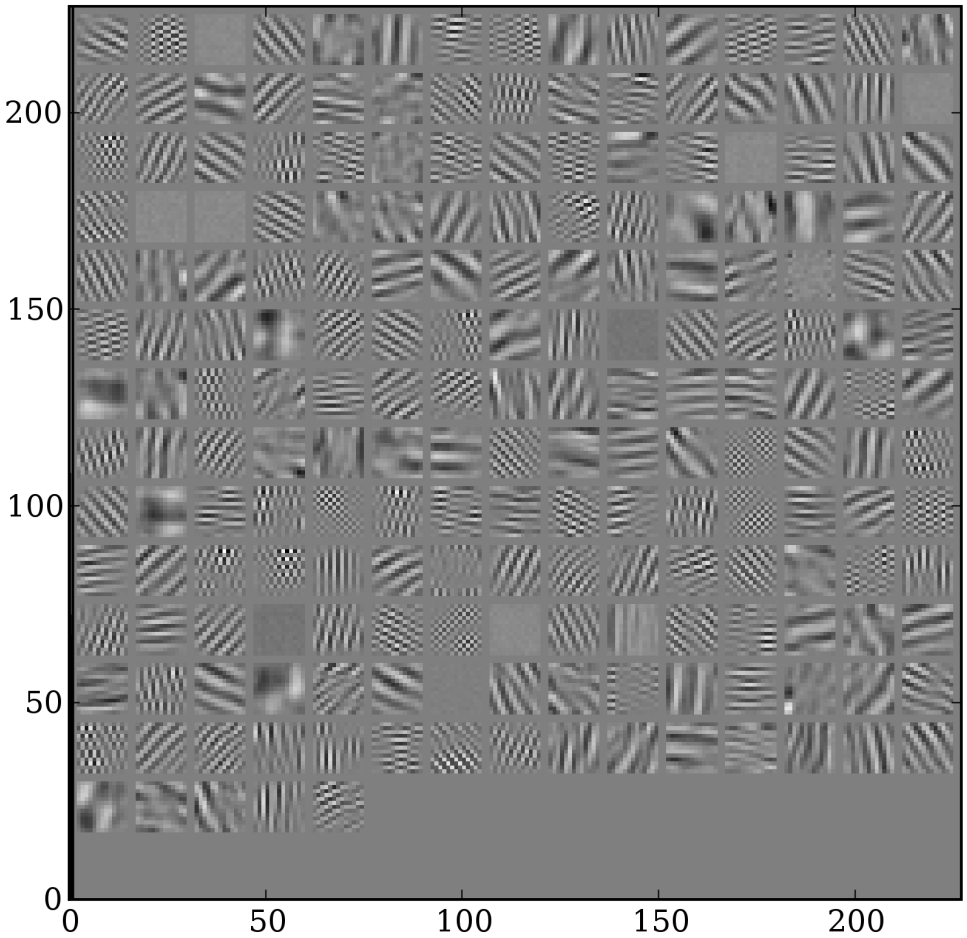
Affine filters



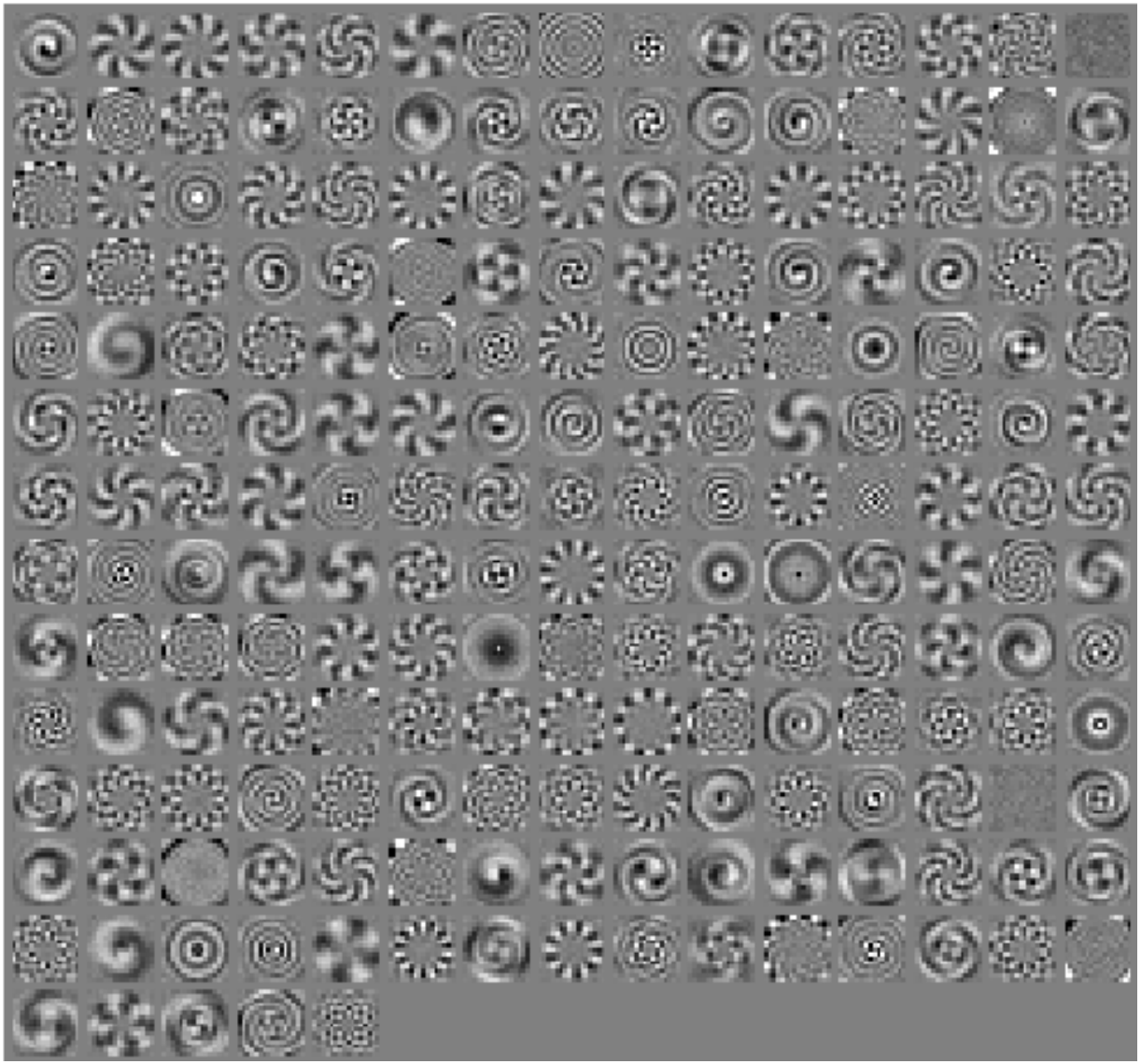
Skaling filters



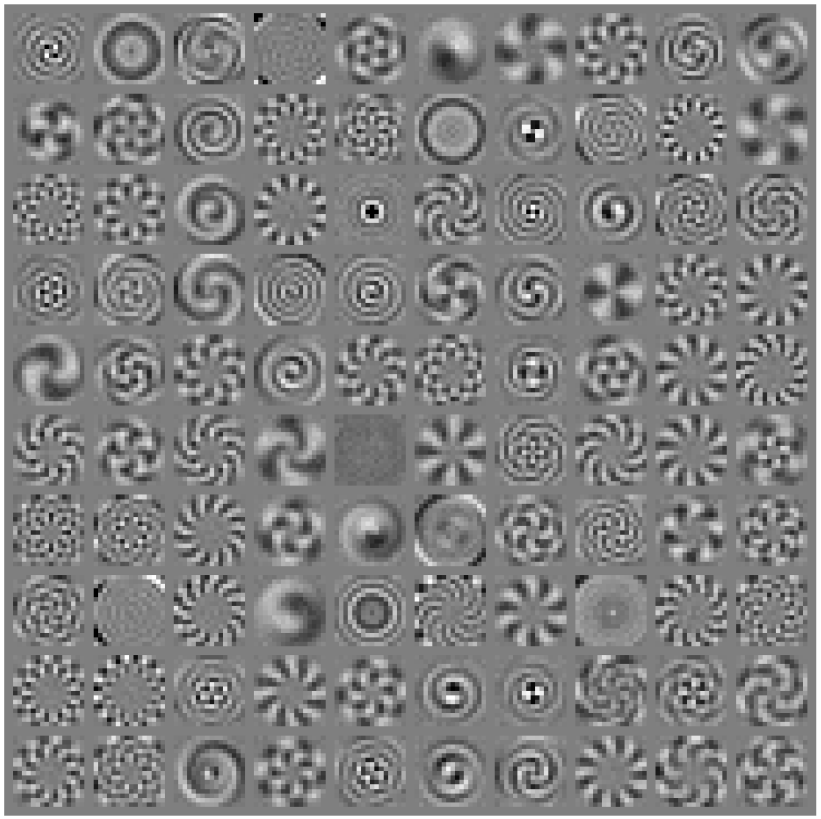
“Filtering” -filters



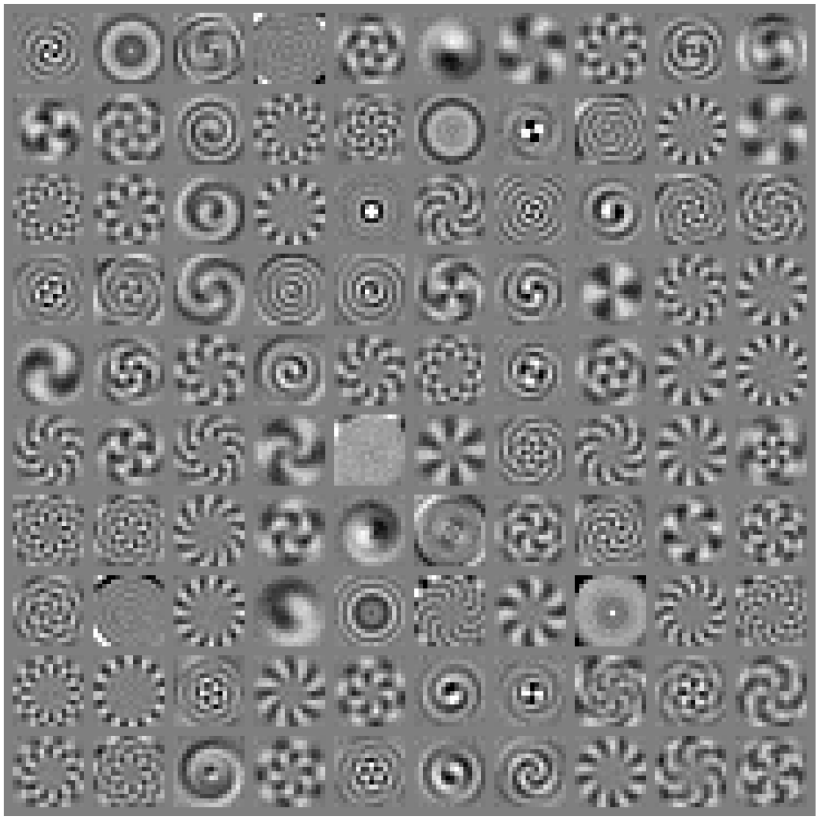
Rotation filters



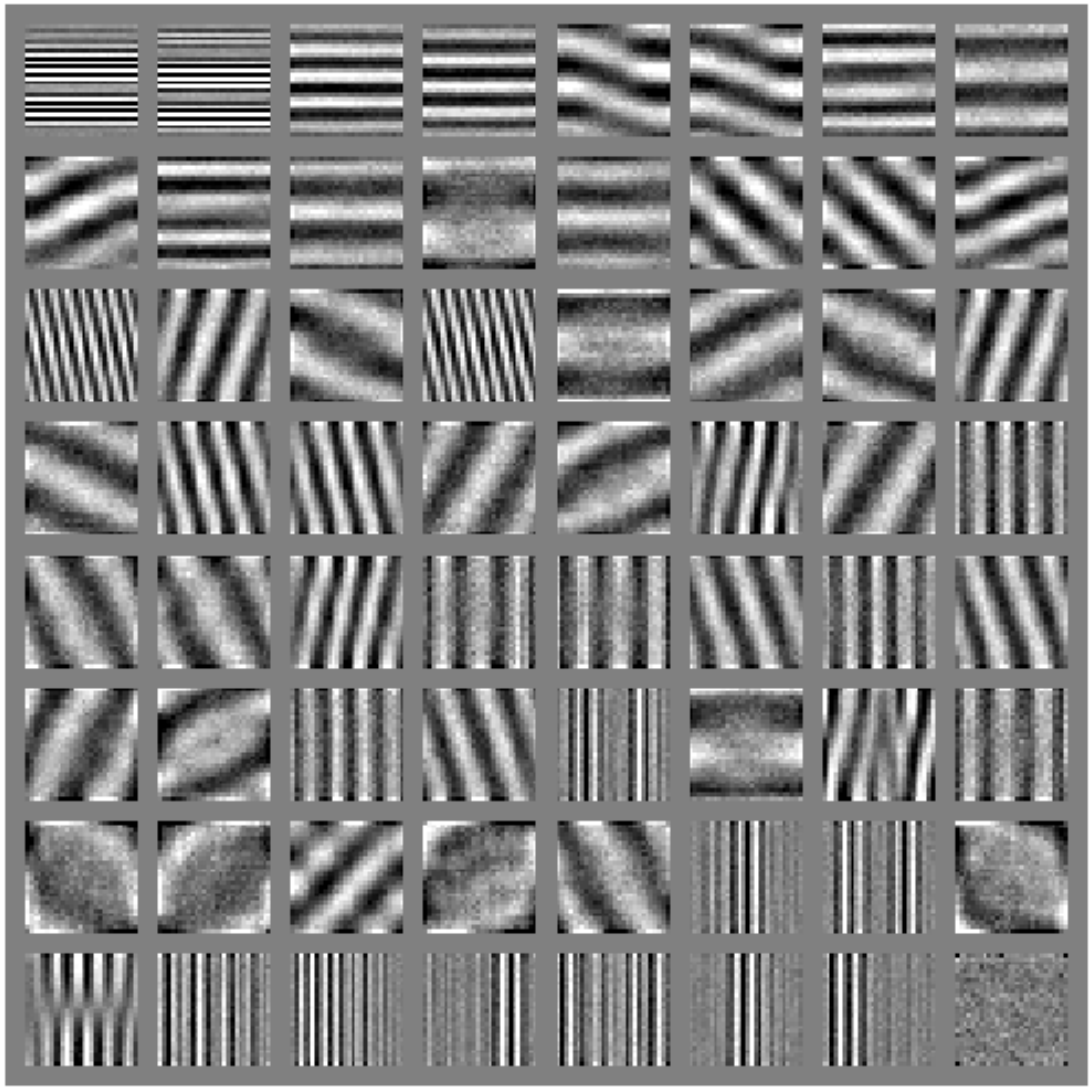
Rotation filters



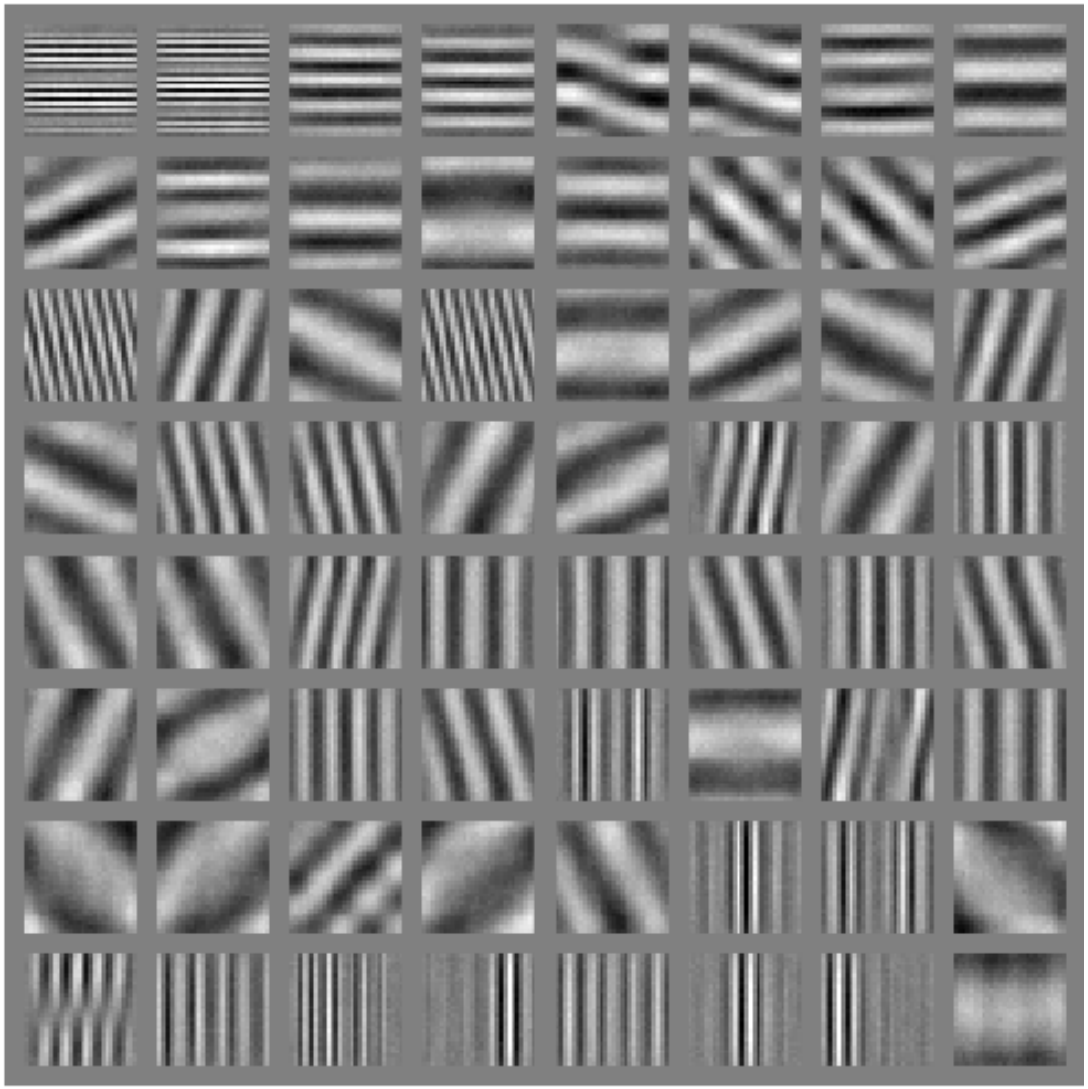
Rotation filters



Filters learned by watching TV



Filters learned by watching TV



Analogy making

$$A : A' \quad :: \quad B : ?$$

Analogy making

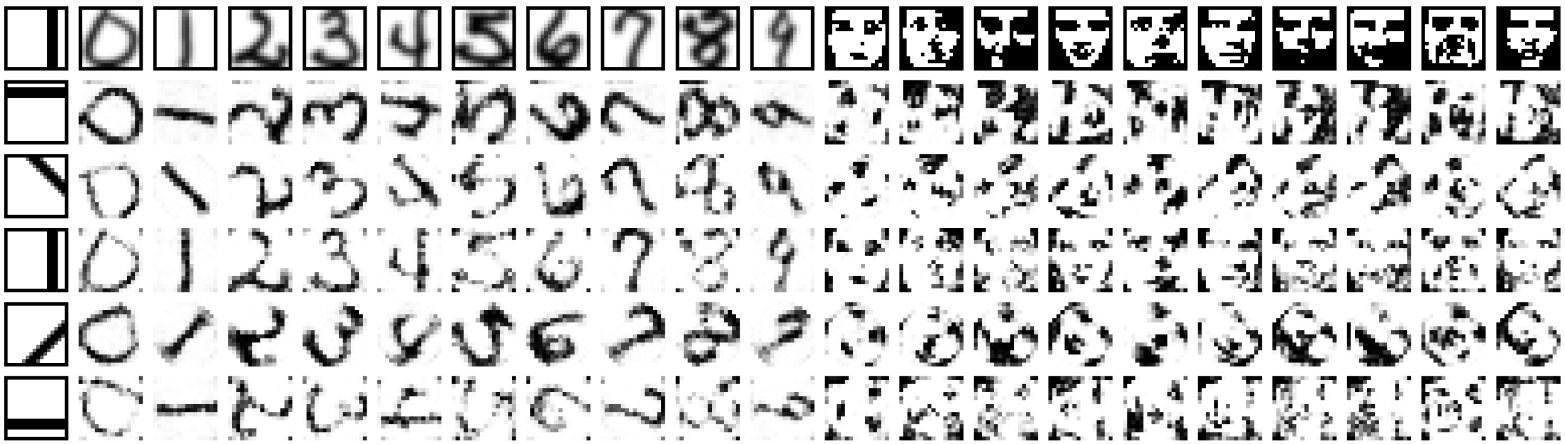
1. Infer transformation from a source image pair:

$$p(\mathbf{z}|\mathbf{x}_{\mathbf{source}}, \mathbf{y}_{\mathbf{source}})$$

2. Apply the transformation to a new image $\mathbf{x}_{\mathbf{target}}$:

$$p(\mathbf{y}|\mathbf{z}, \mathbf{x}_{\mathbf{target}})$$

Analogy making



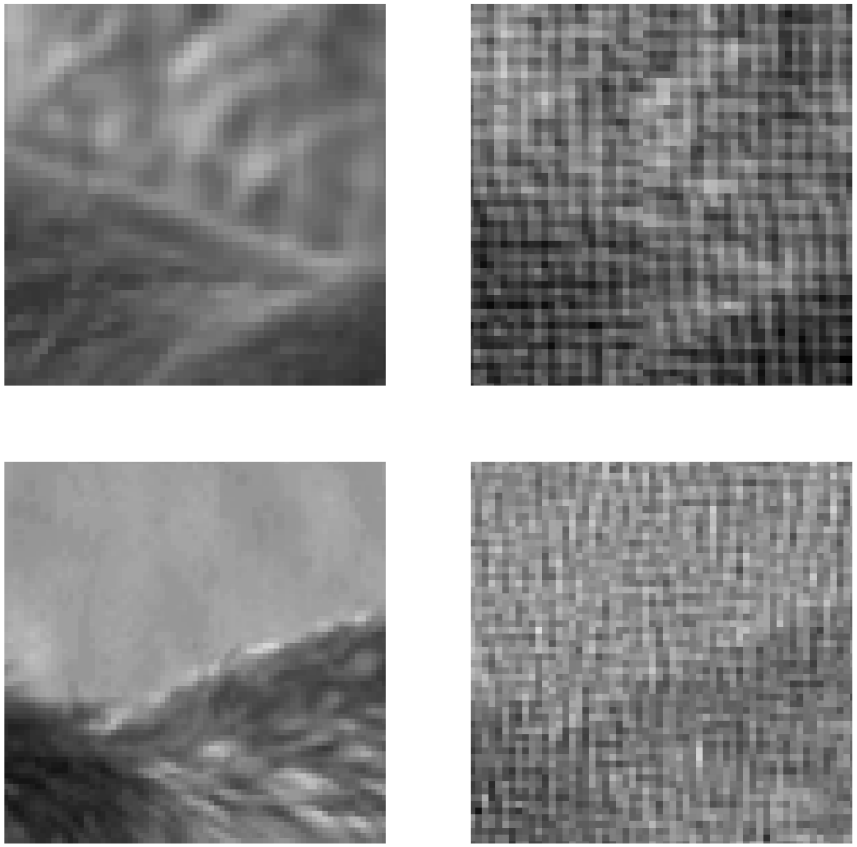
Analogy making (II)



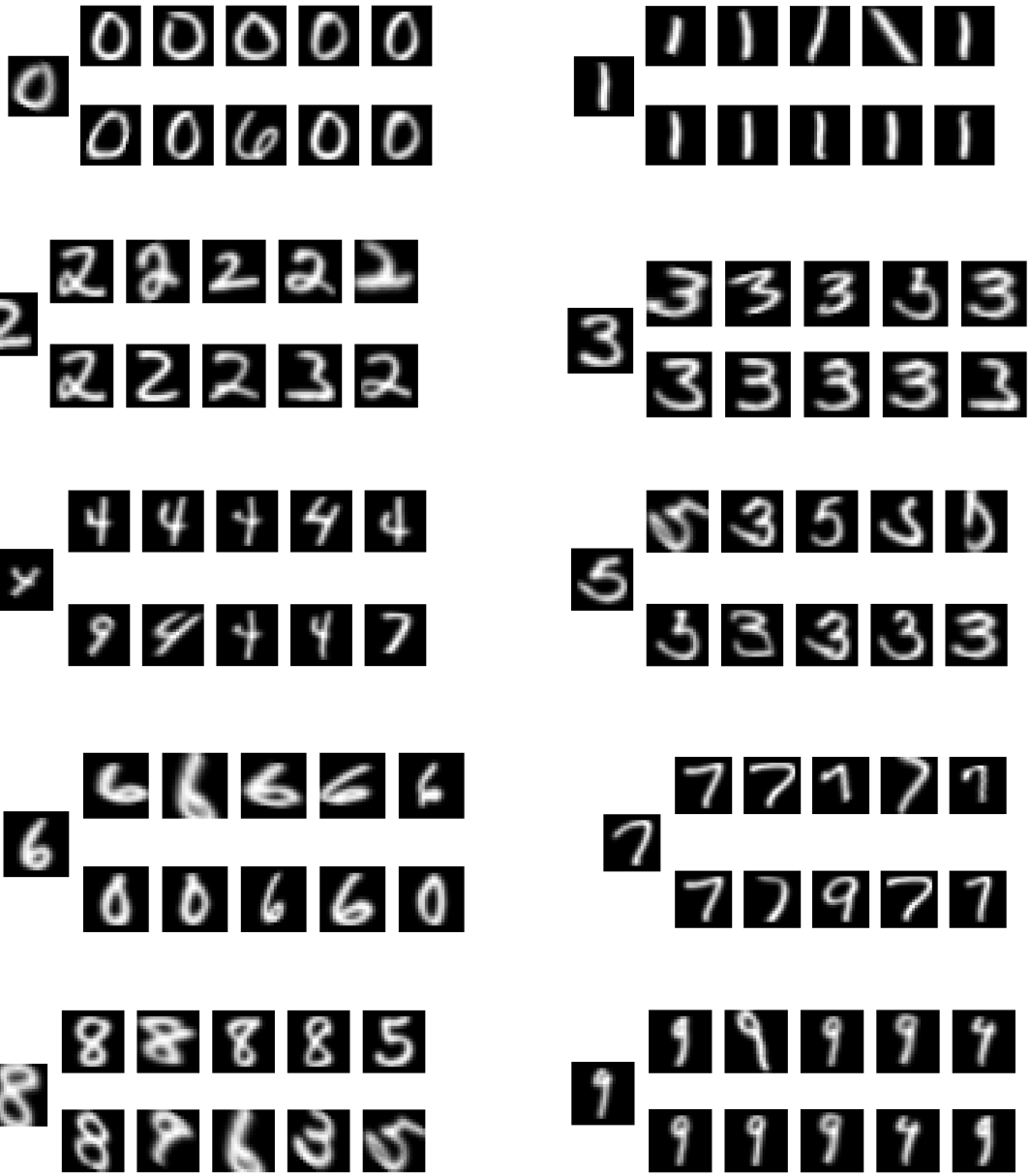



- (Convolutional)

Analogy making (II)



An invariant metric



Face transformations

- Filters learned from faces:



Facial expression transfer

- Imitate the expression; *retain the identity*

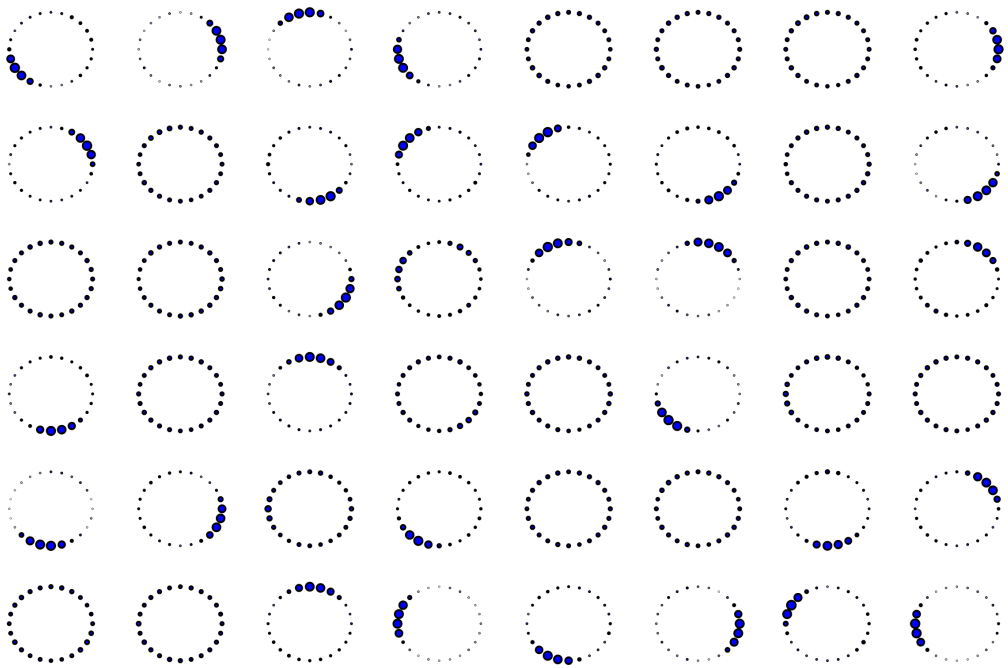


Face matching

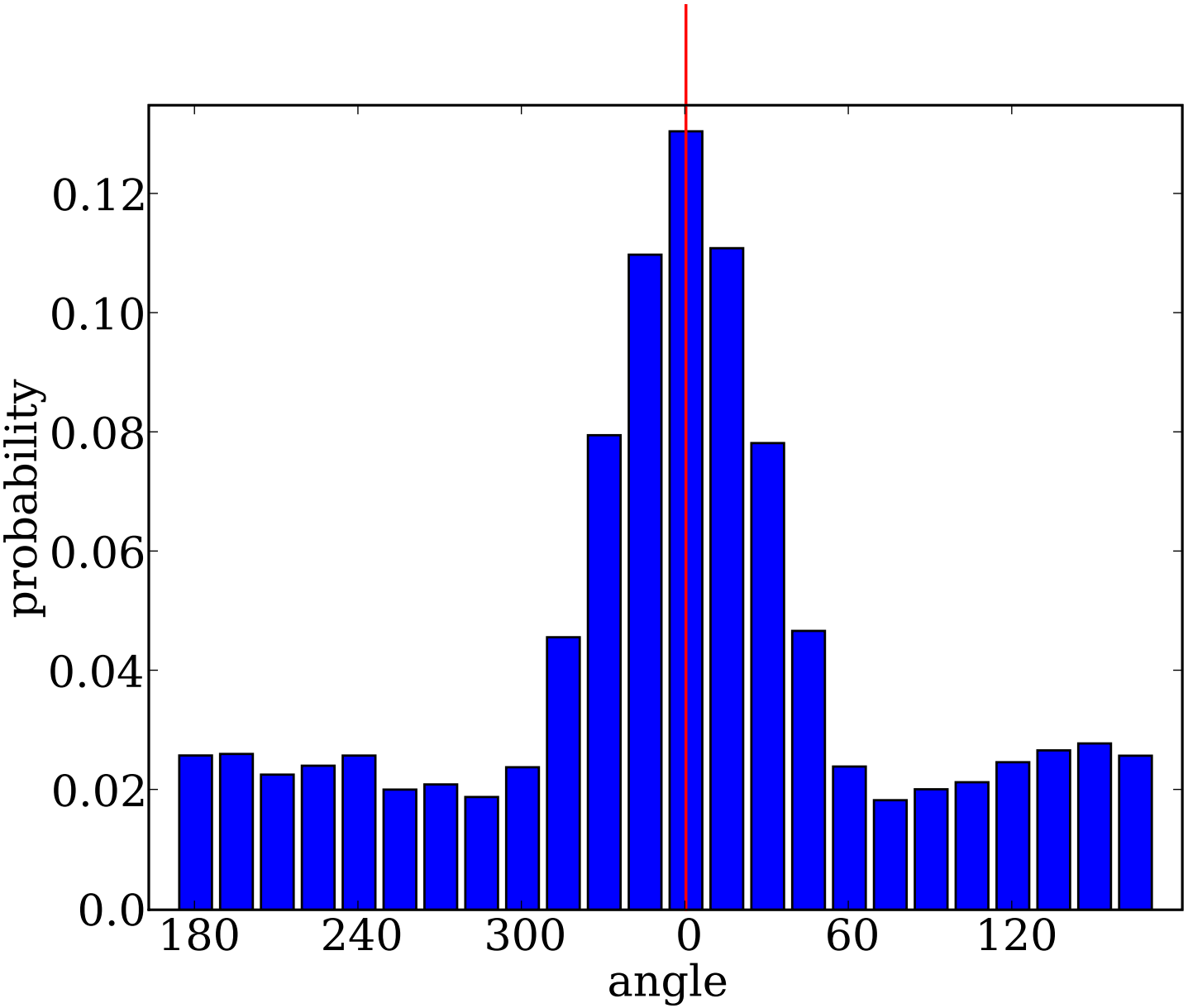
model/Task	TFD ID	TFD Exp	PUBFIG ID	AFFINE
cosine	0.848	0.663	0.649	0.721
rBM	0.869	0.656	0.647	0.799
conditional	0.805	0.634	0.557	0.825
bilinear	0.905	0.637	**0.774**	0.812
3-way	0.932	**0.705**	0.771	0.930
3-way symm	**0.951**	0.695	0.762	**0.931**

Stacking

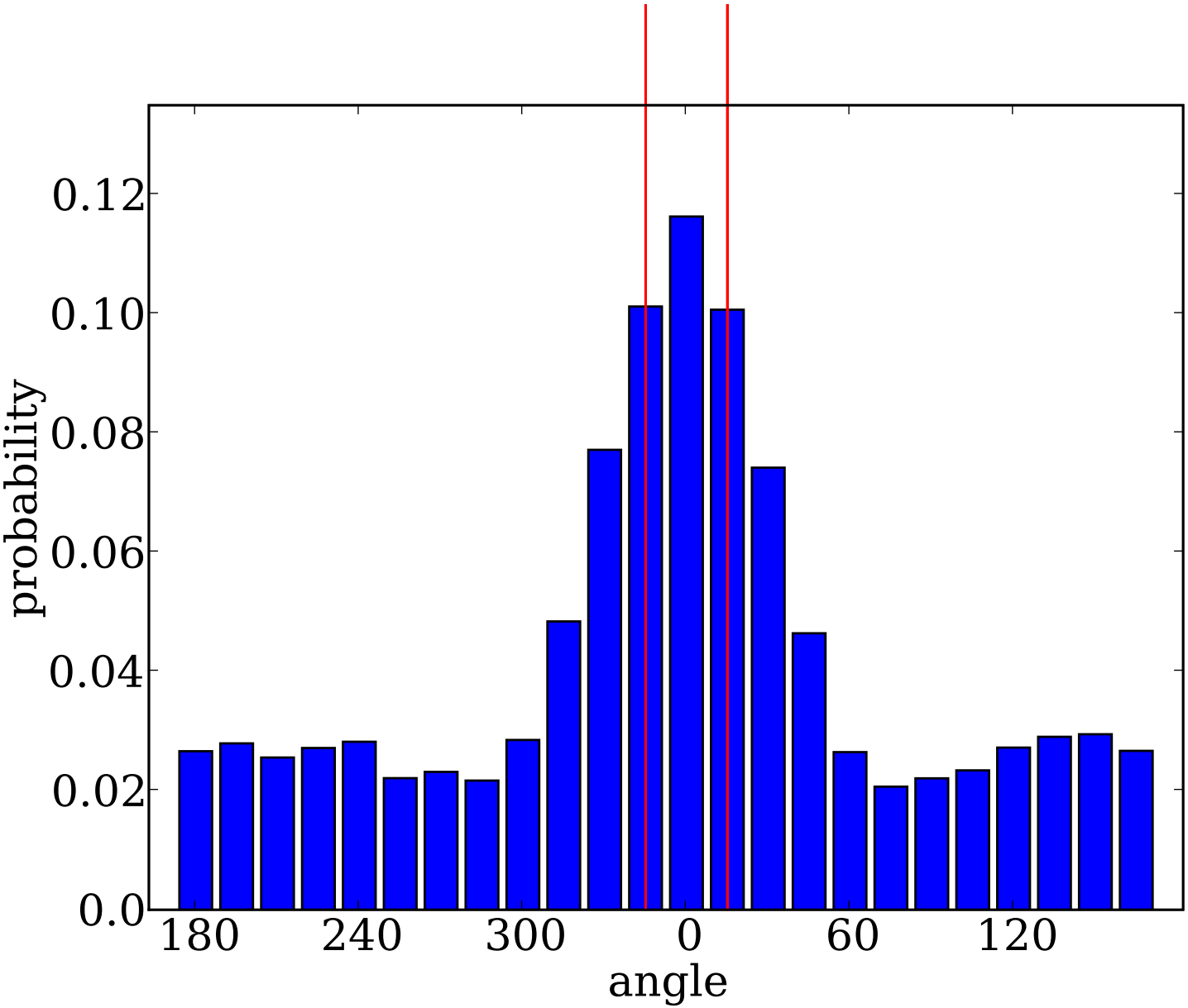
- Hiddens form a population code.
- To get more interpretable features, define a higher, sparse layer of latent variables.
- Then measure, how strongly each of the higher variables responds to patterns moving in different directions:



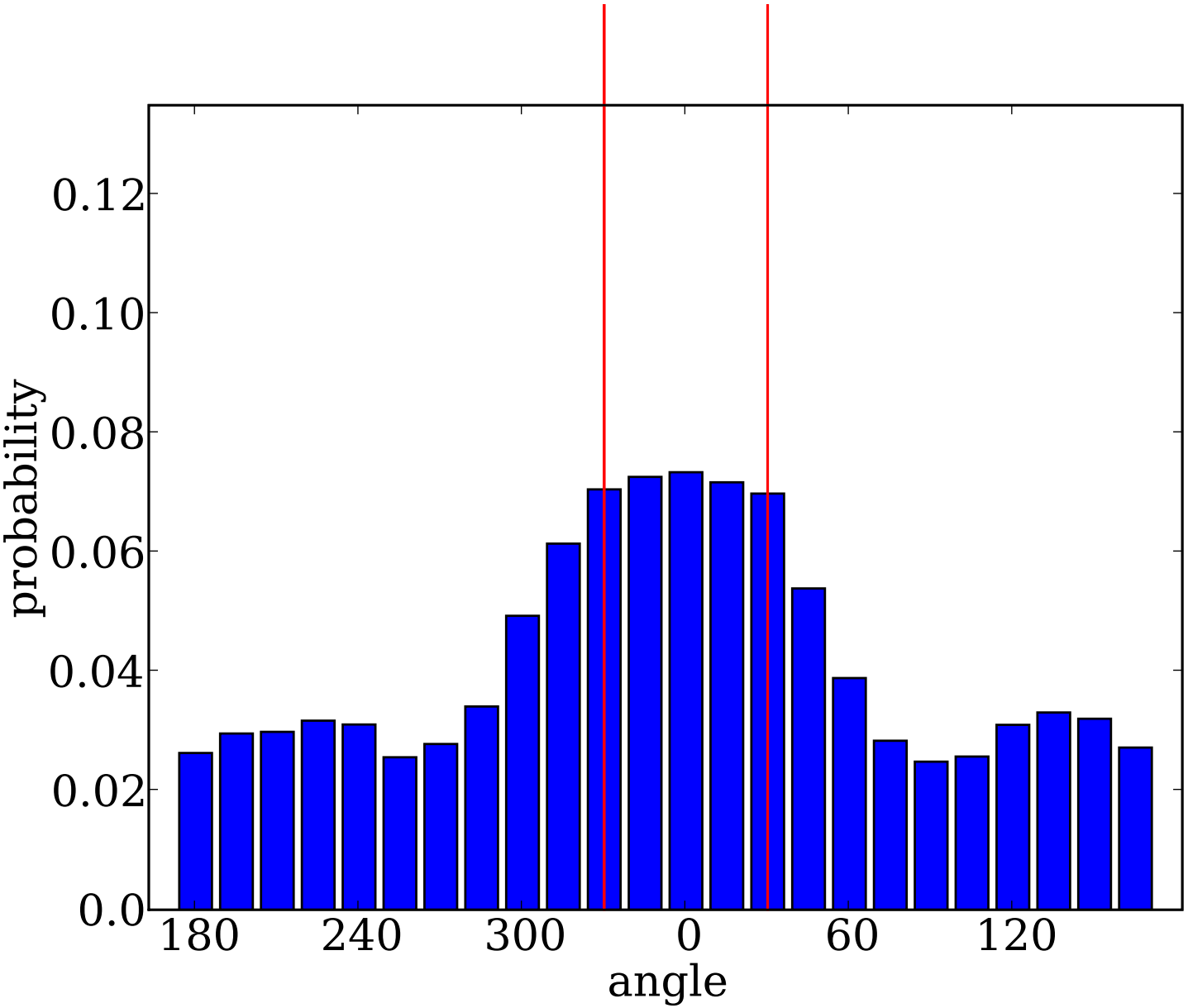
Transparent motion



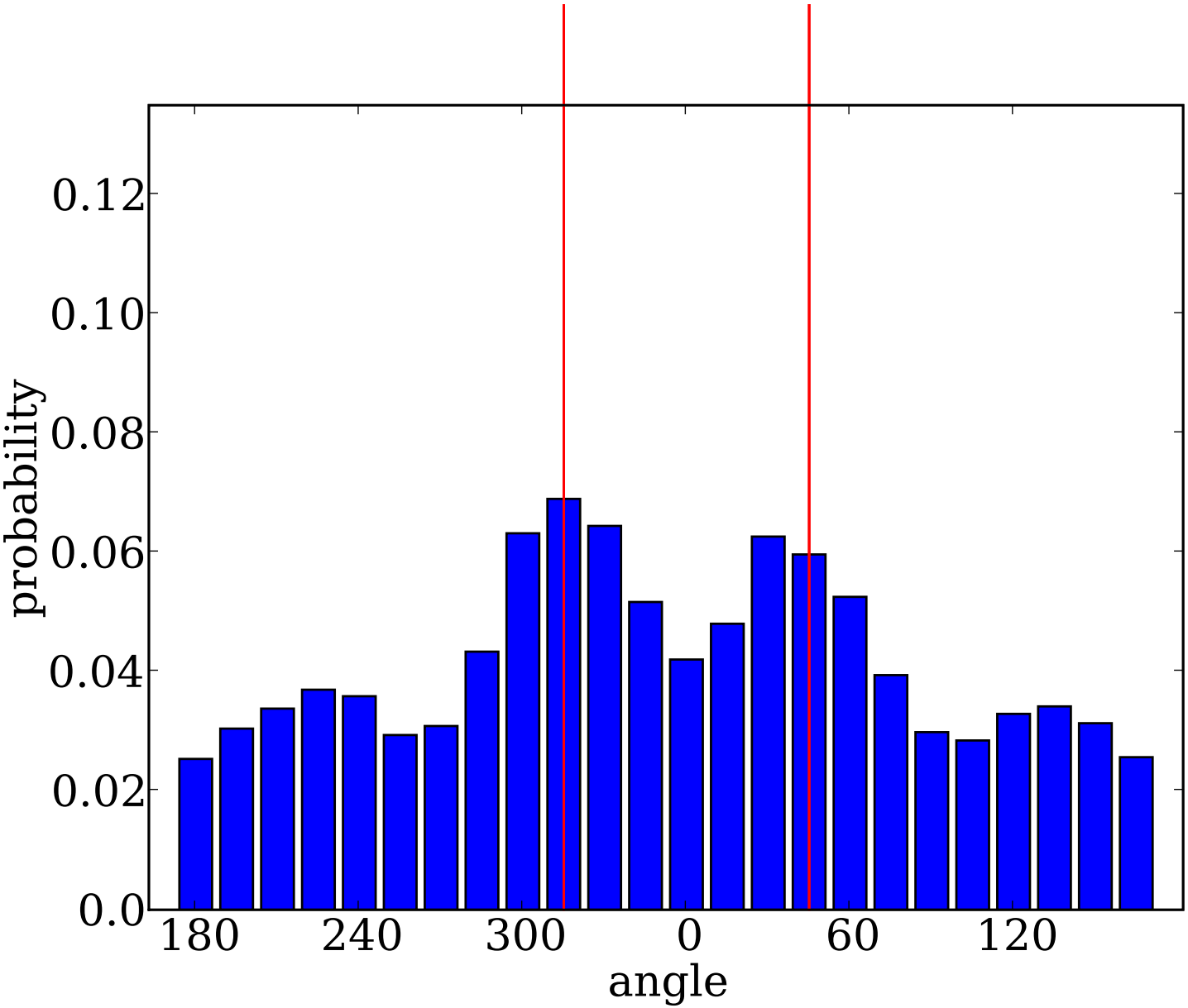
Transparent motion



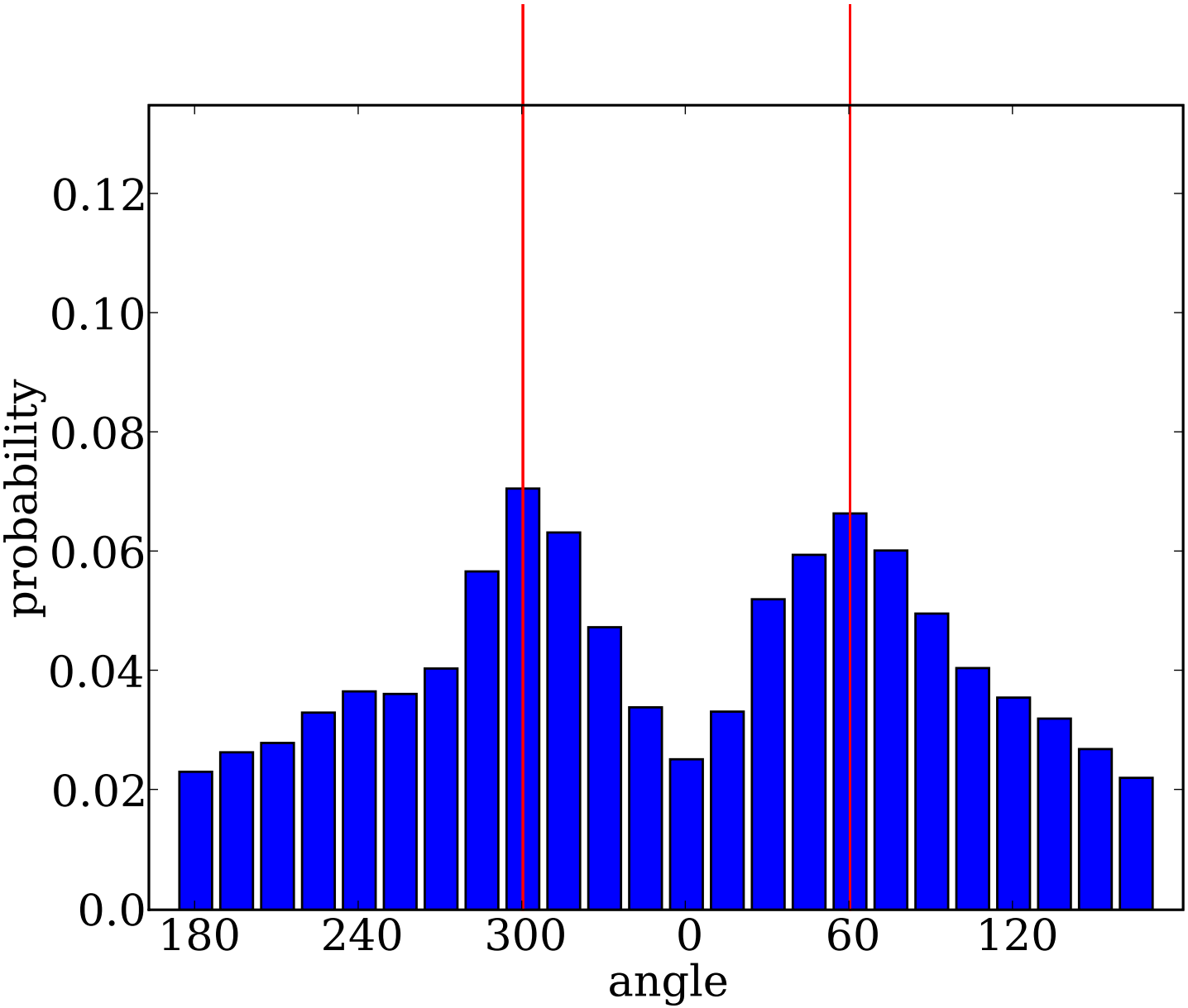
Transparent motion



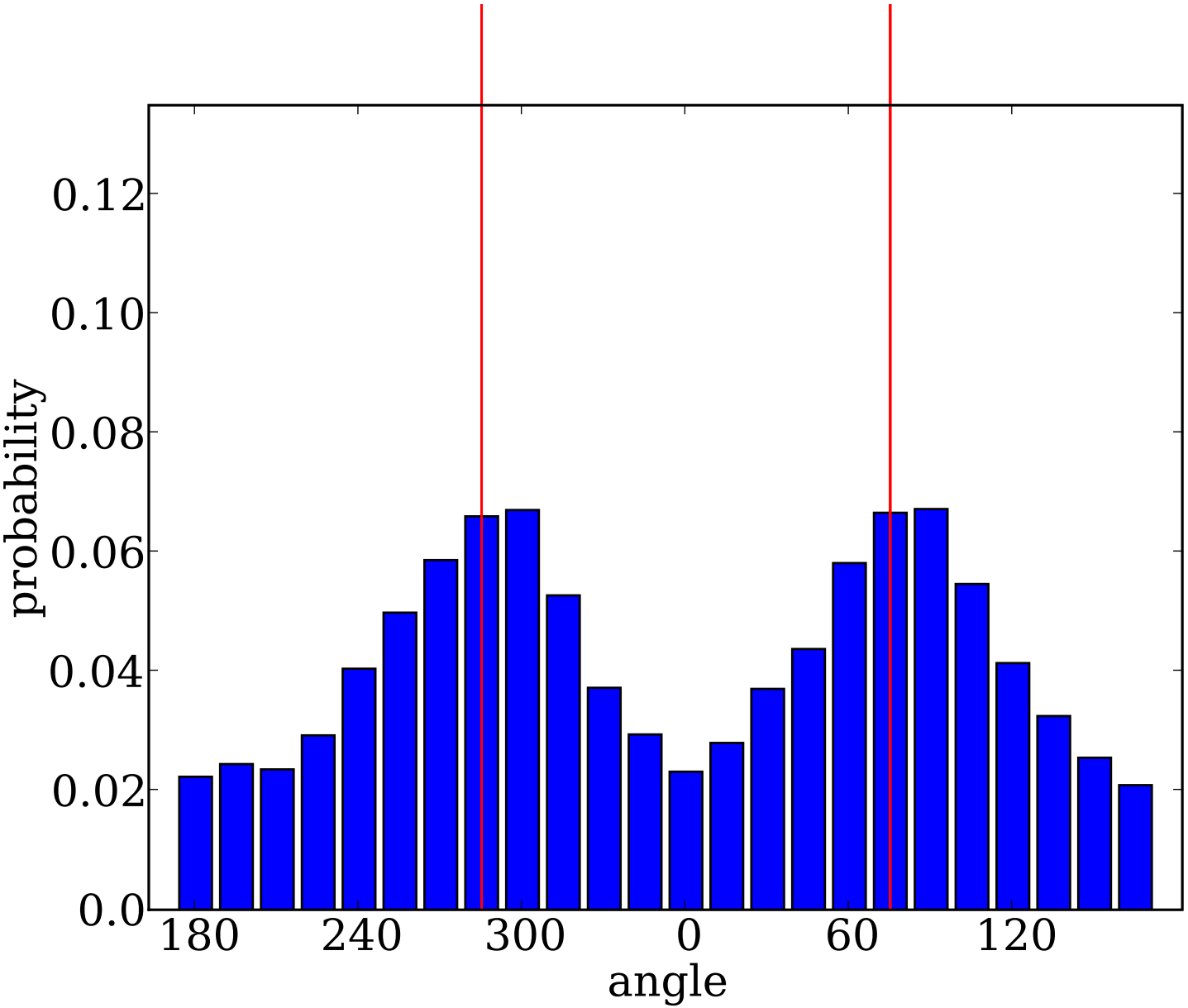
Transparent motion



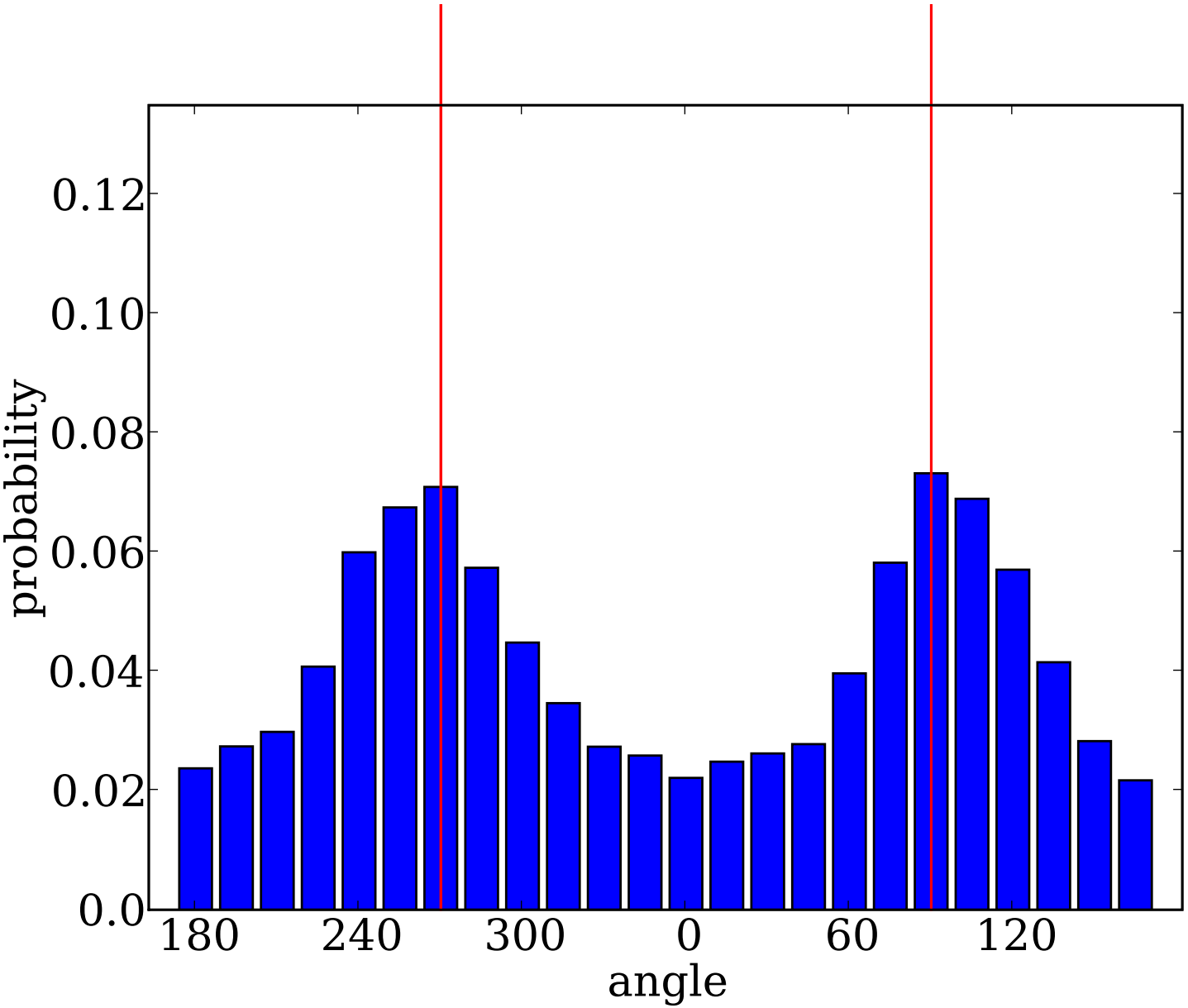
Transparent motion



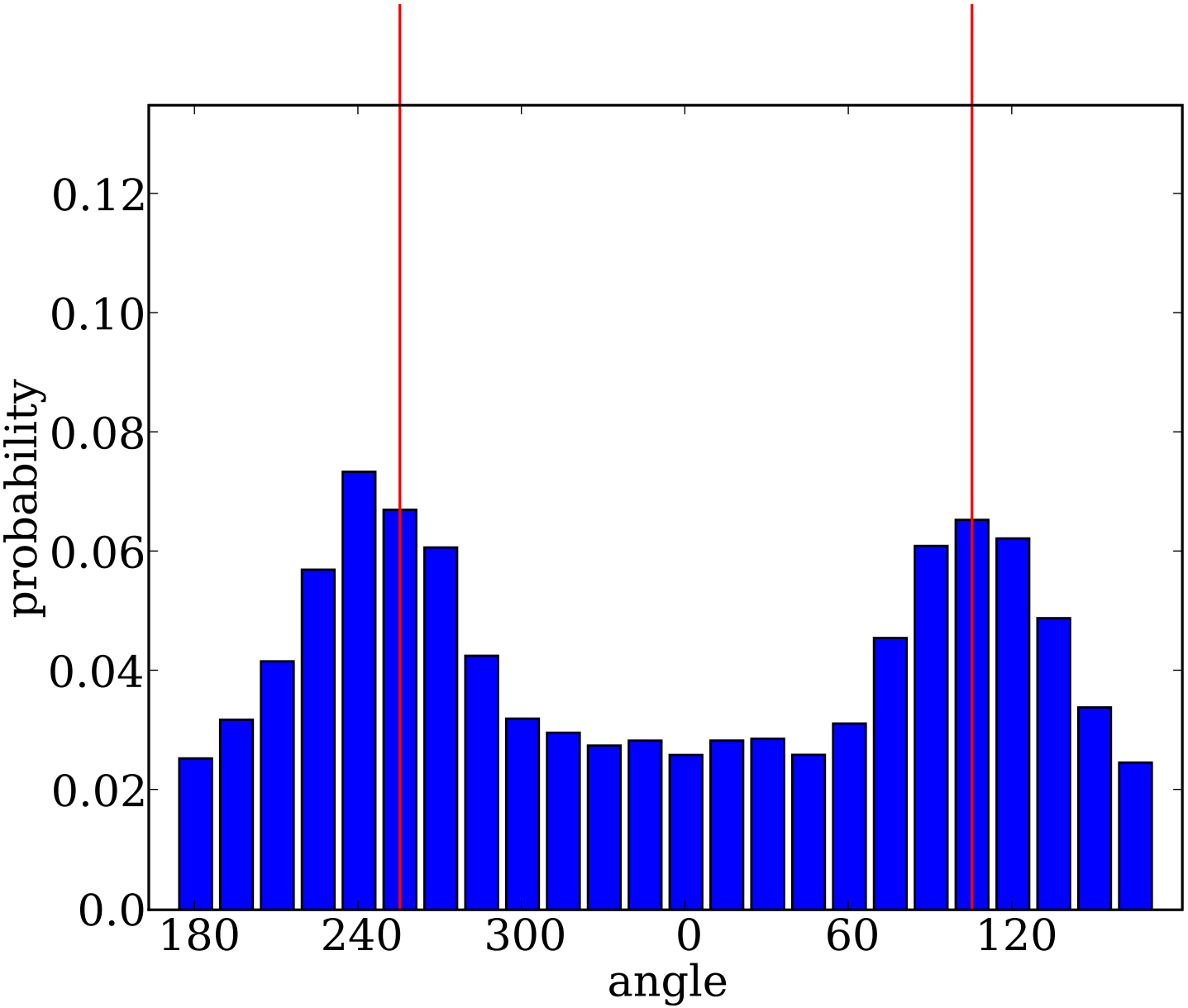
Transparent motion



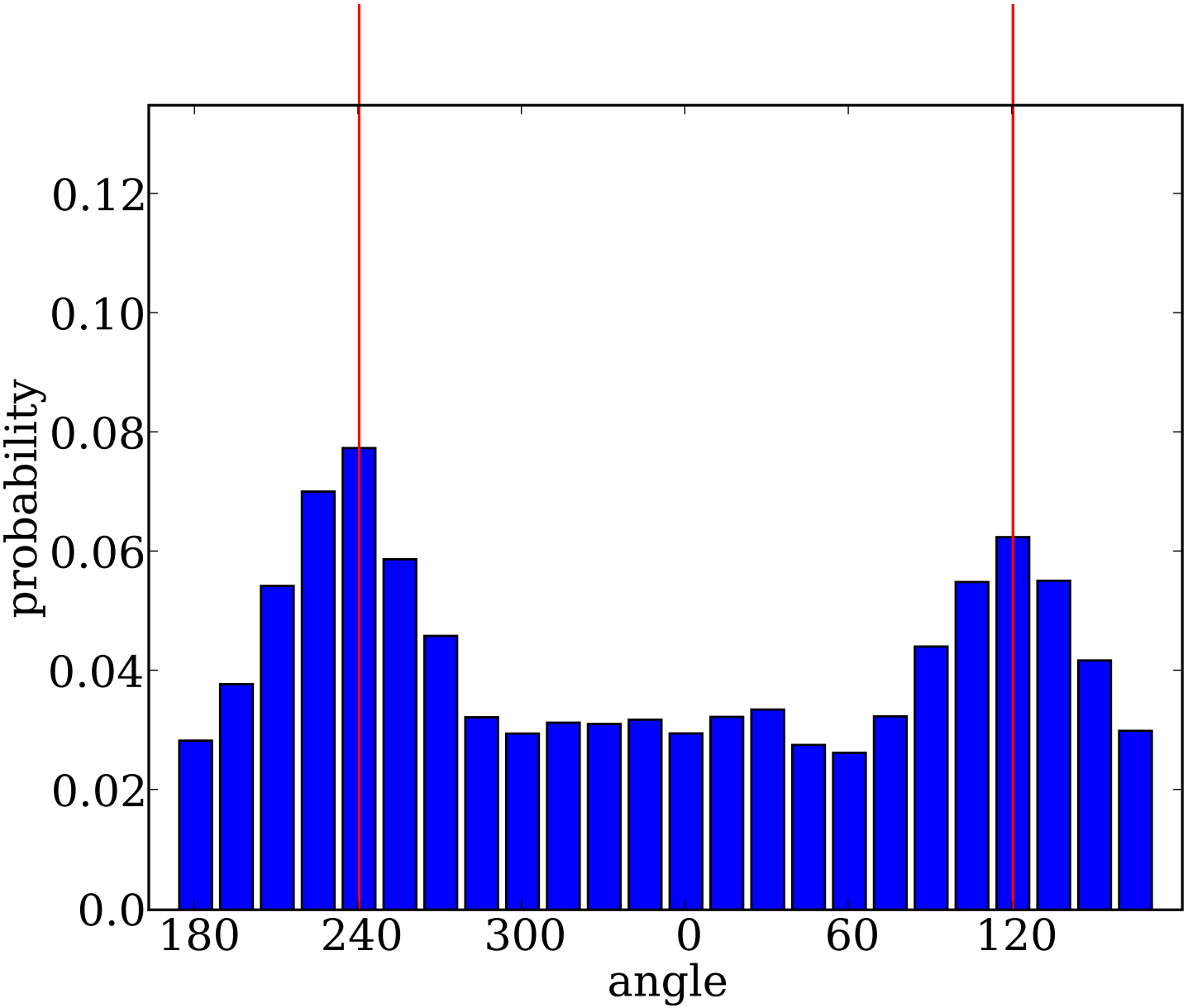
Transparent motion



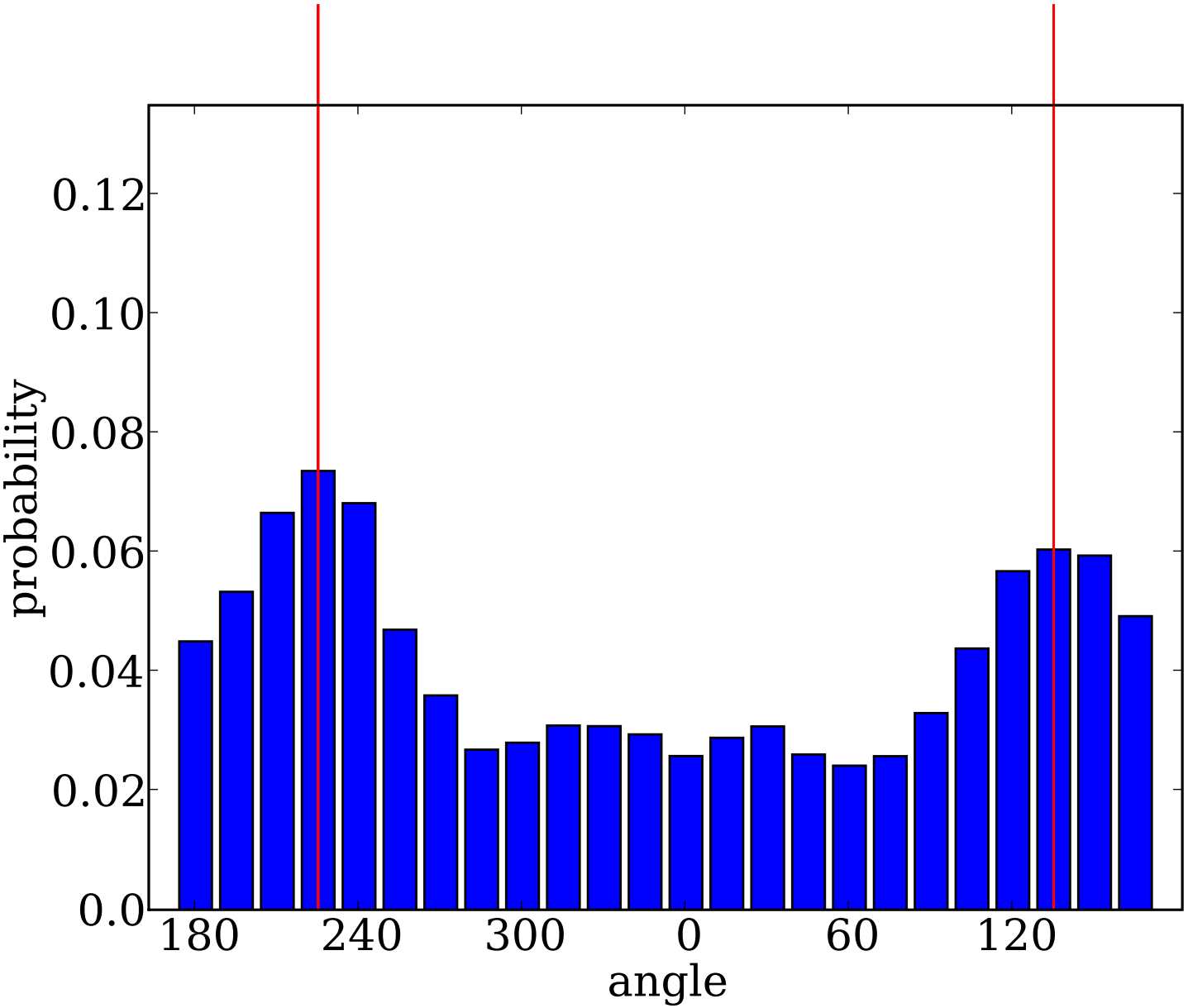
Transparent motion



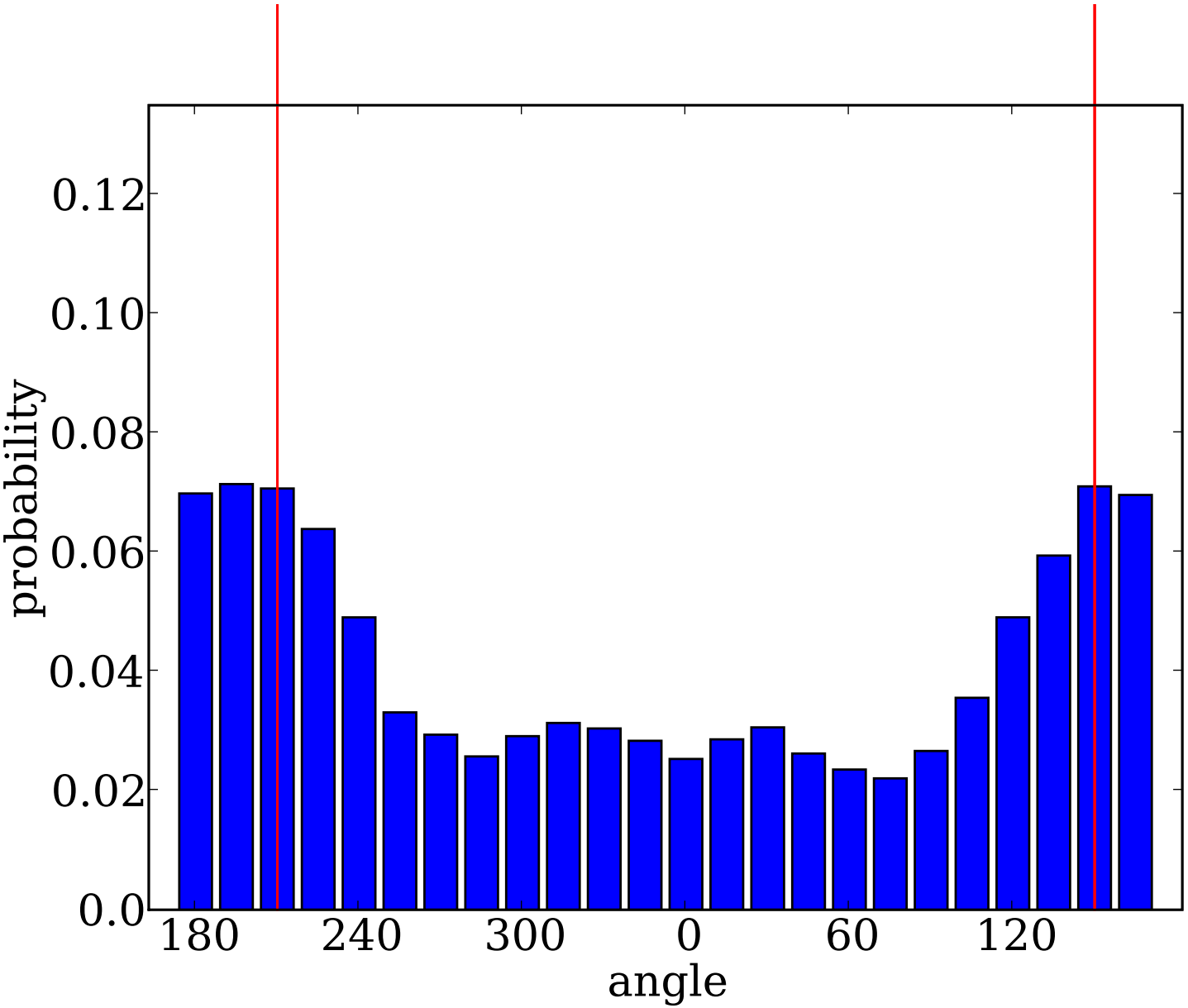
Transparent motion



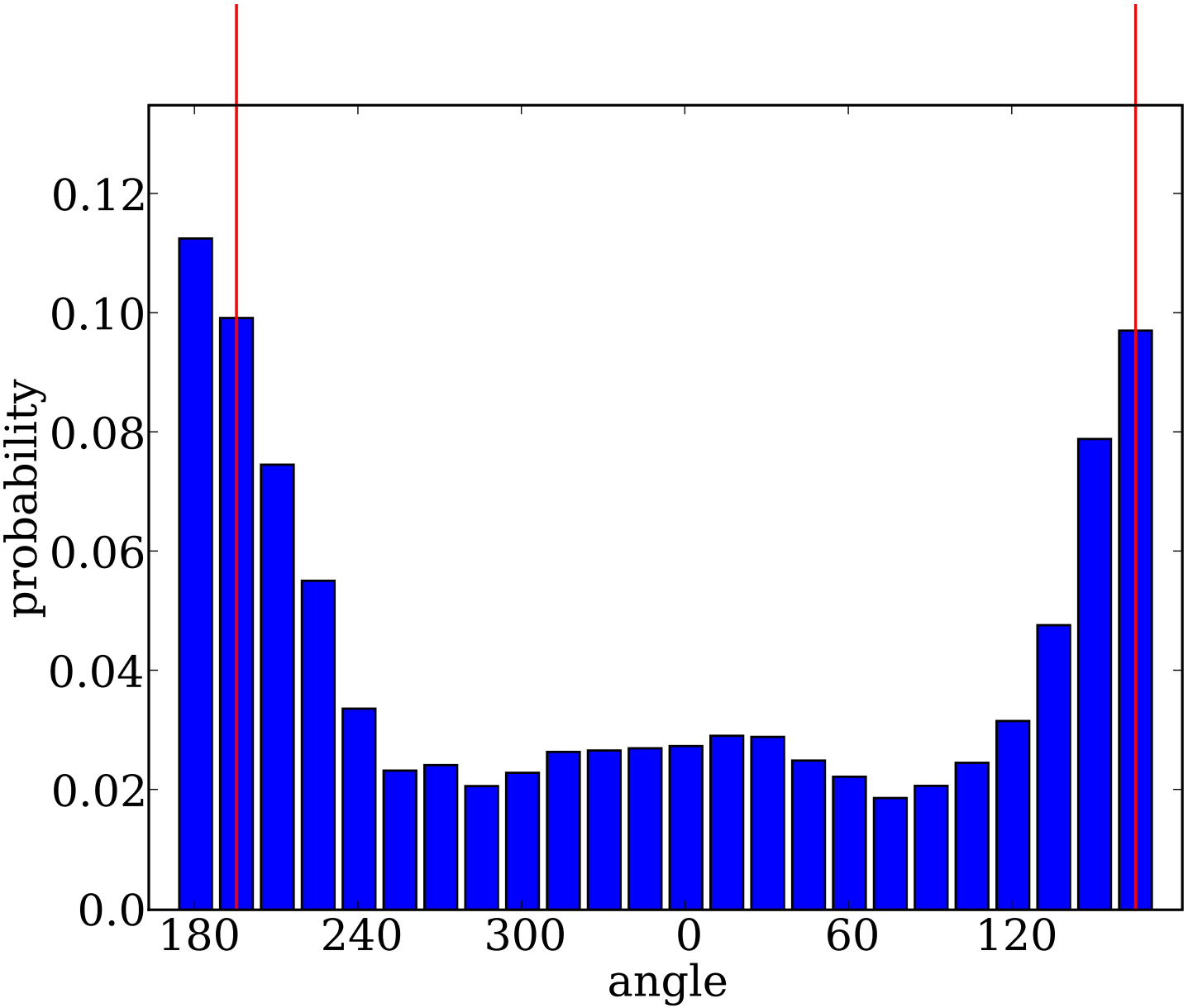
Transparent motion



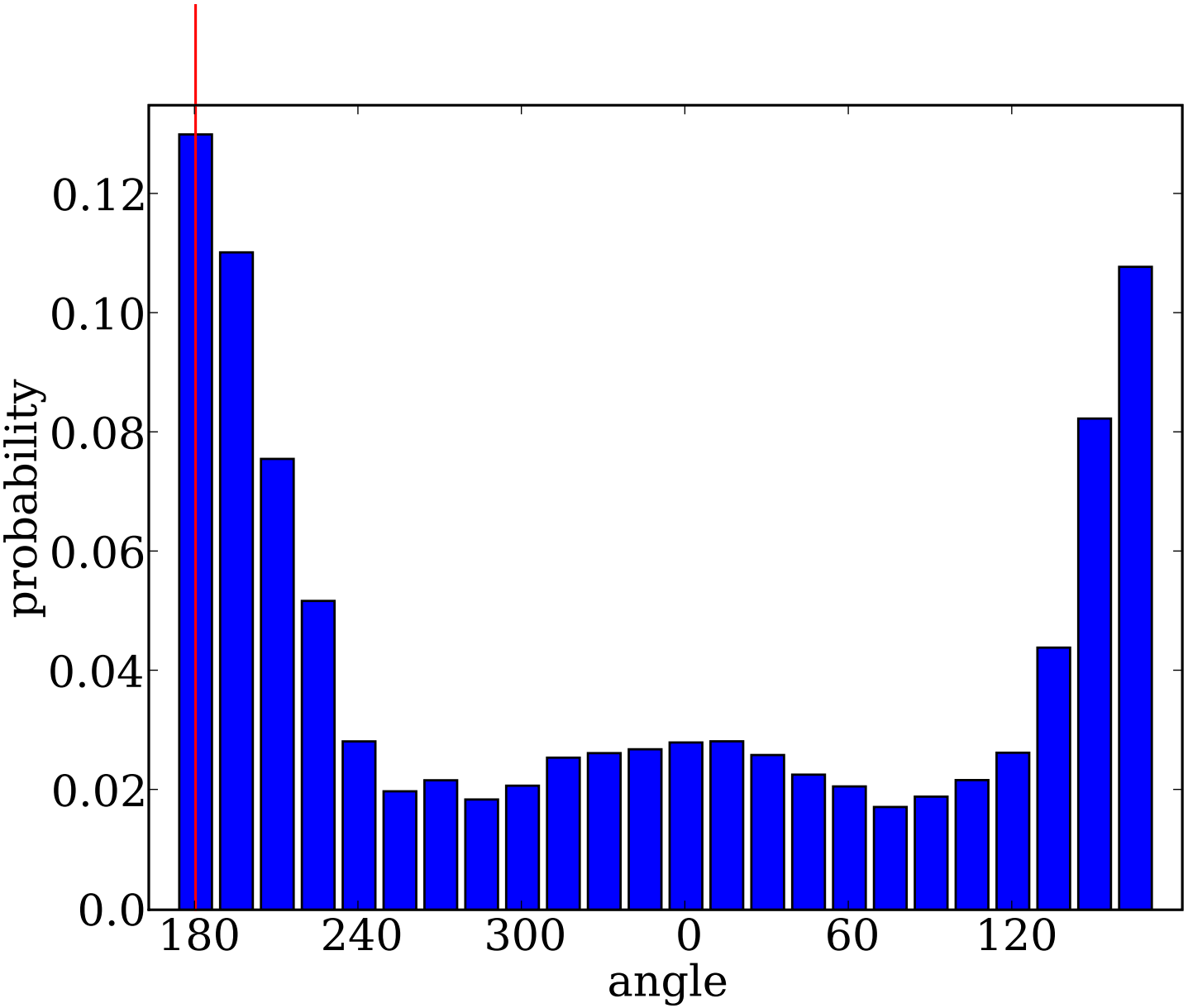
Transparent motion



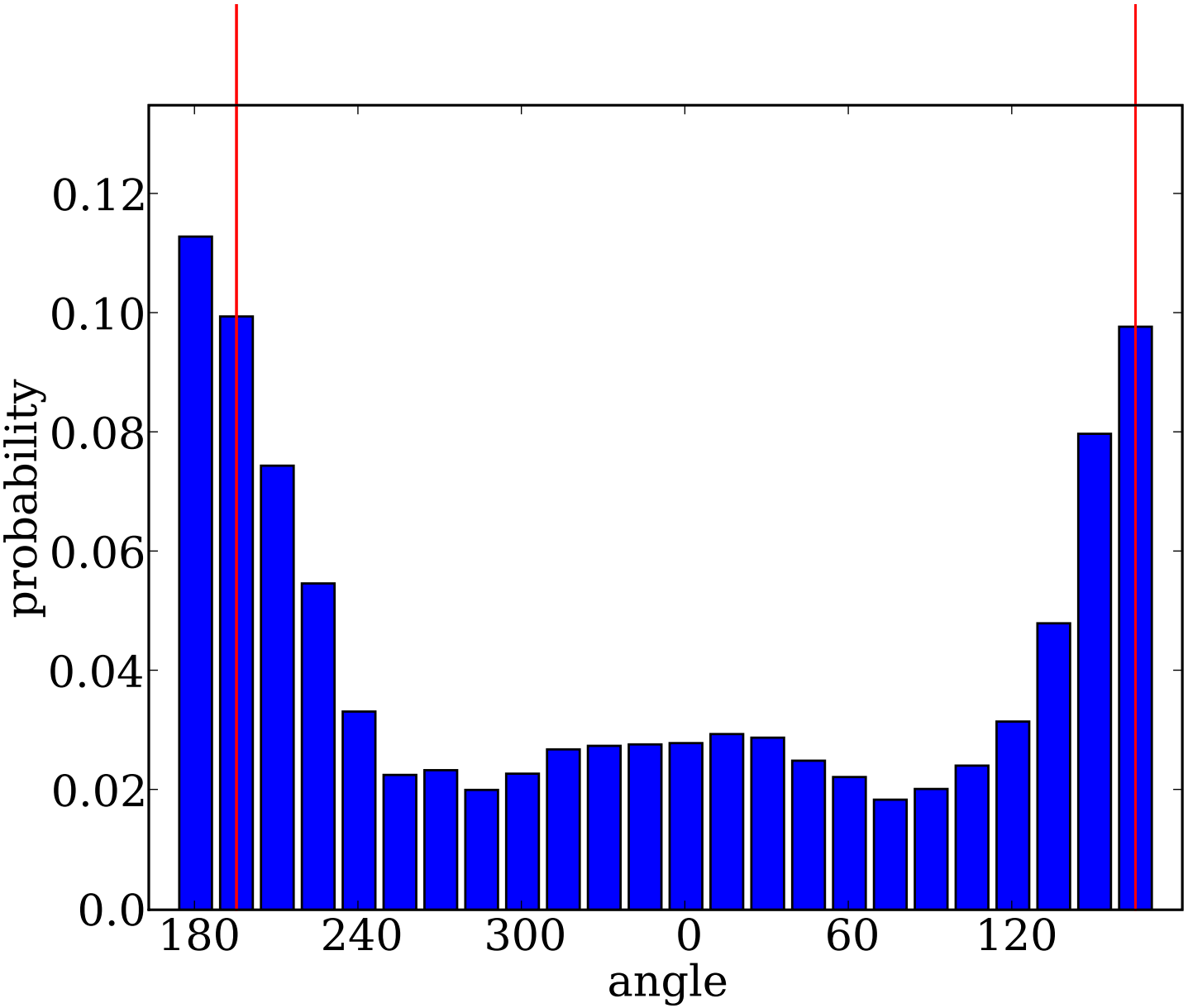
Transparent motion



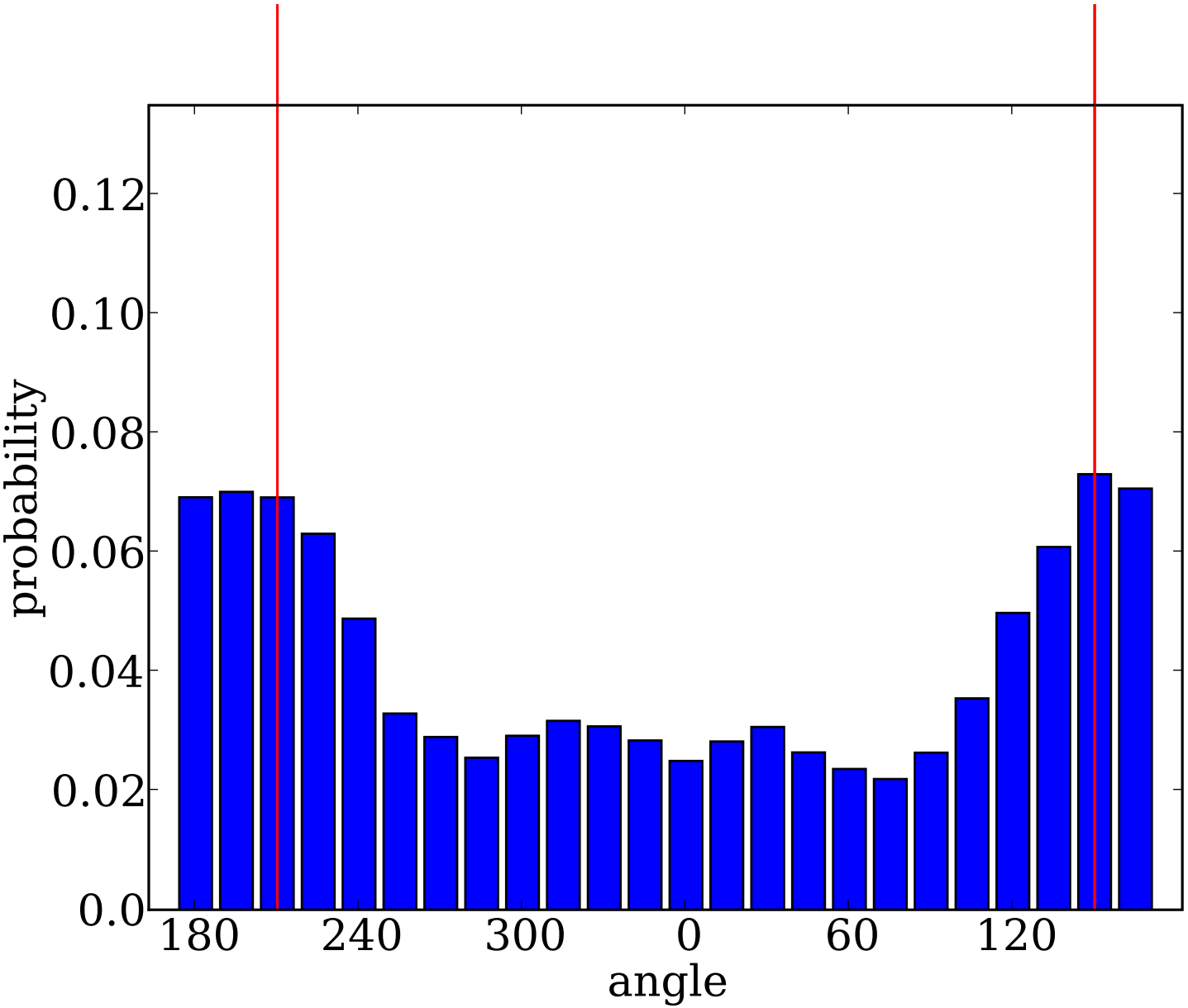
Transparent motion



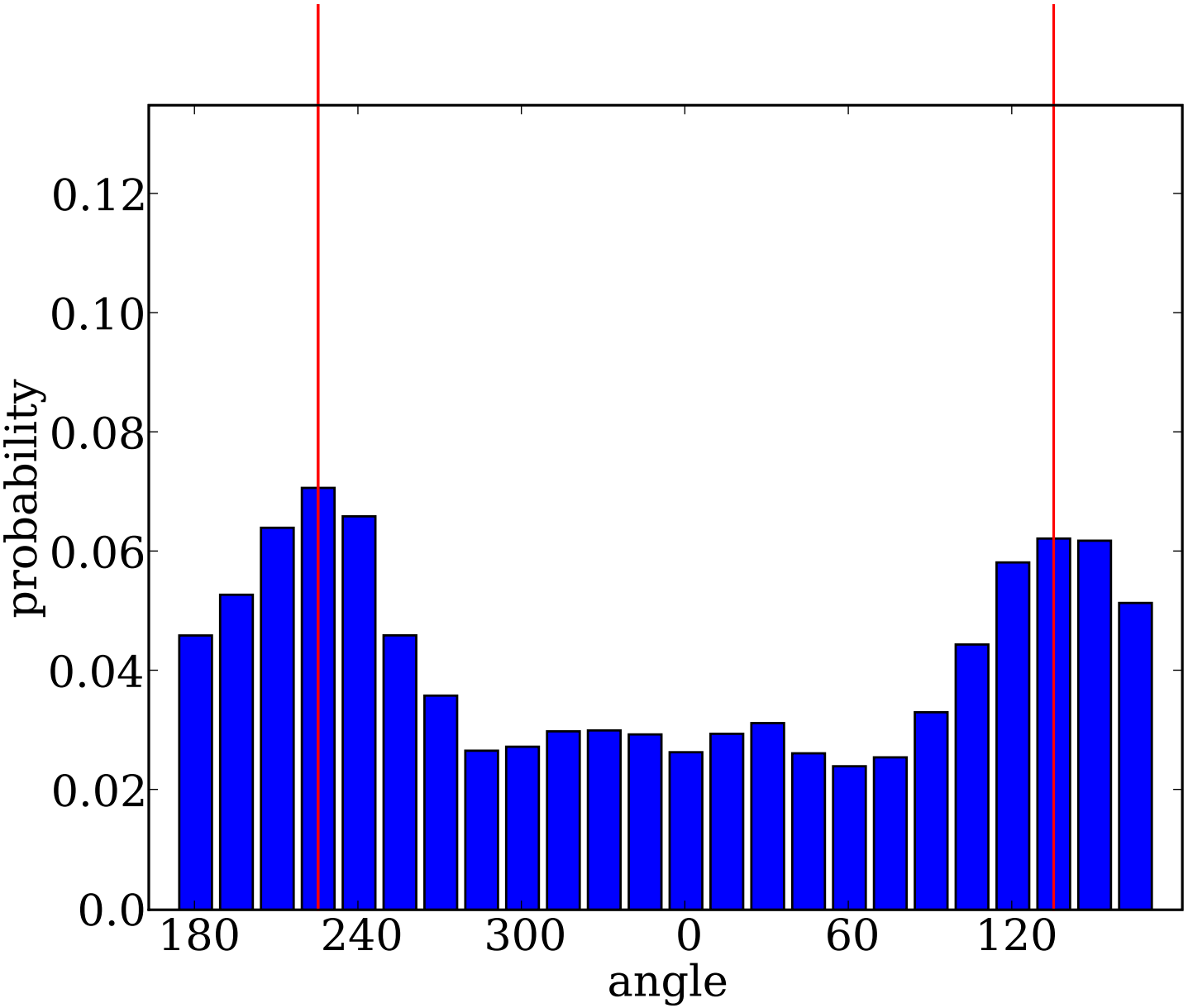
Transparent motion



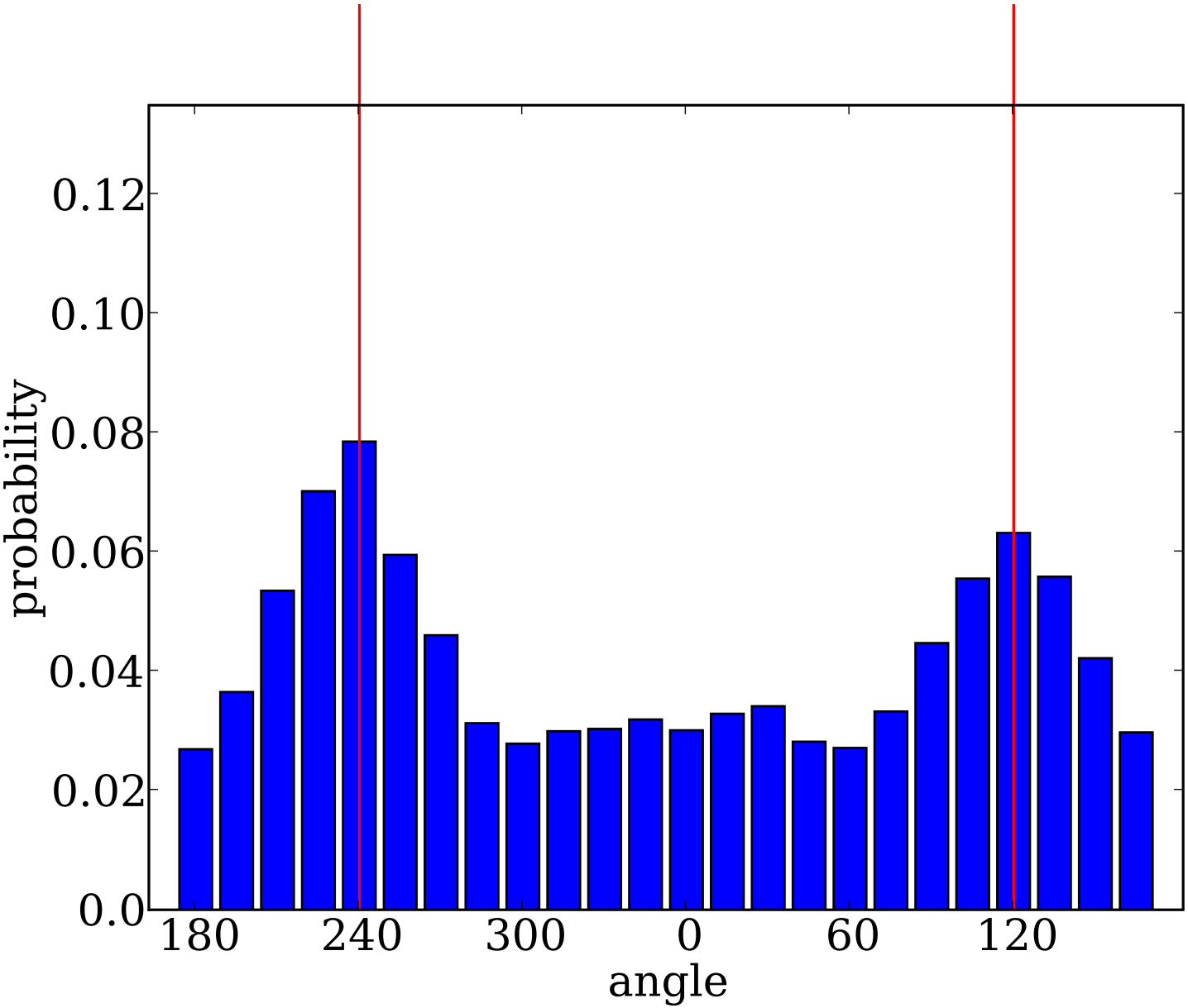
Transparent motion



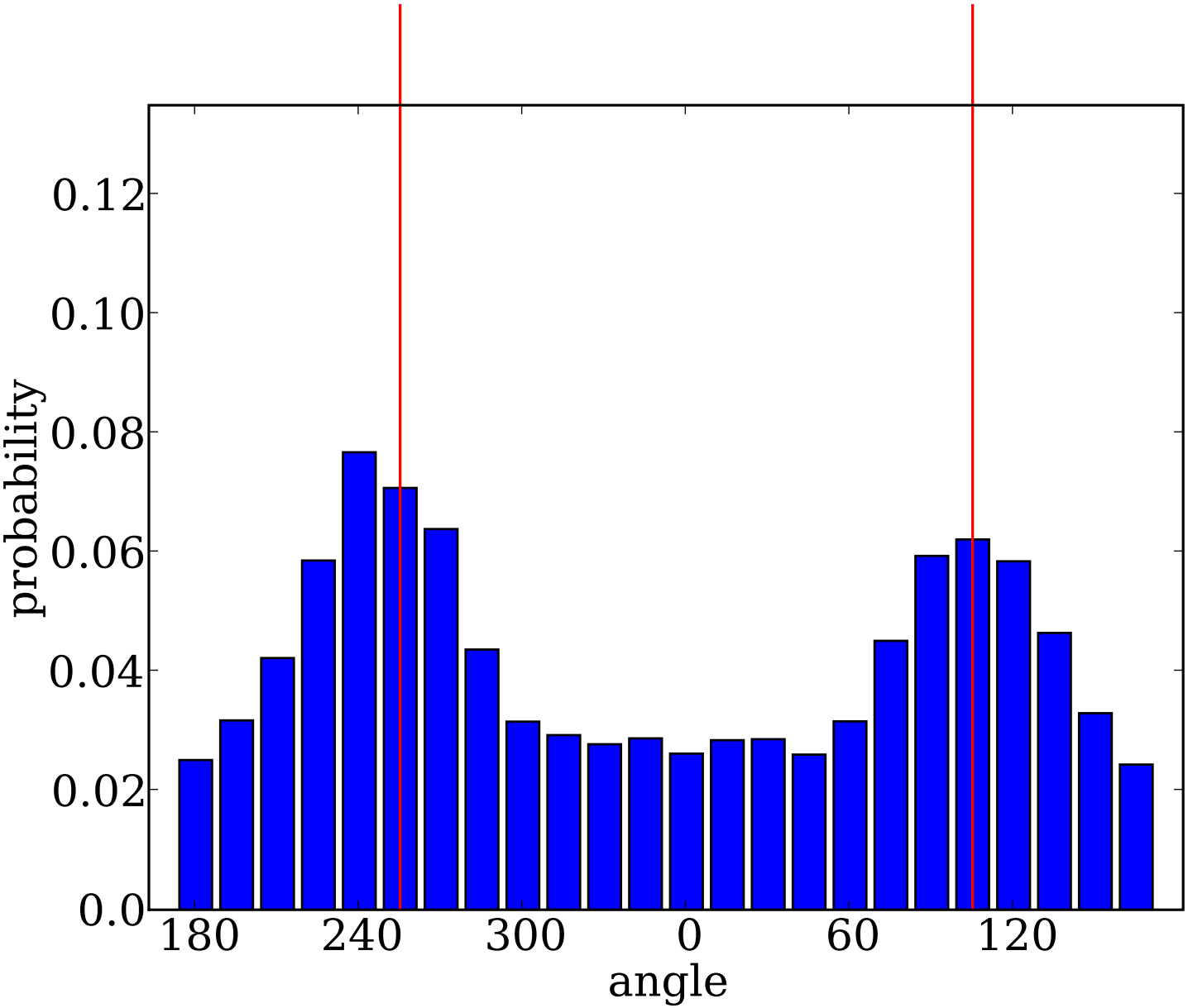
Transparent motion



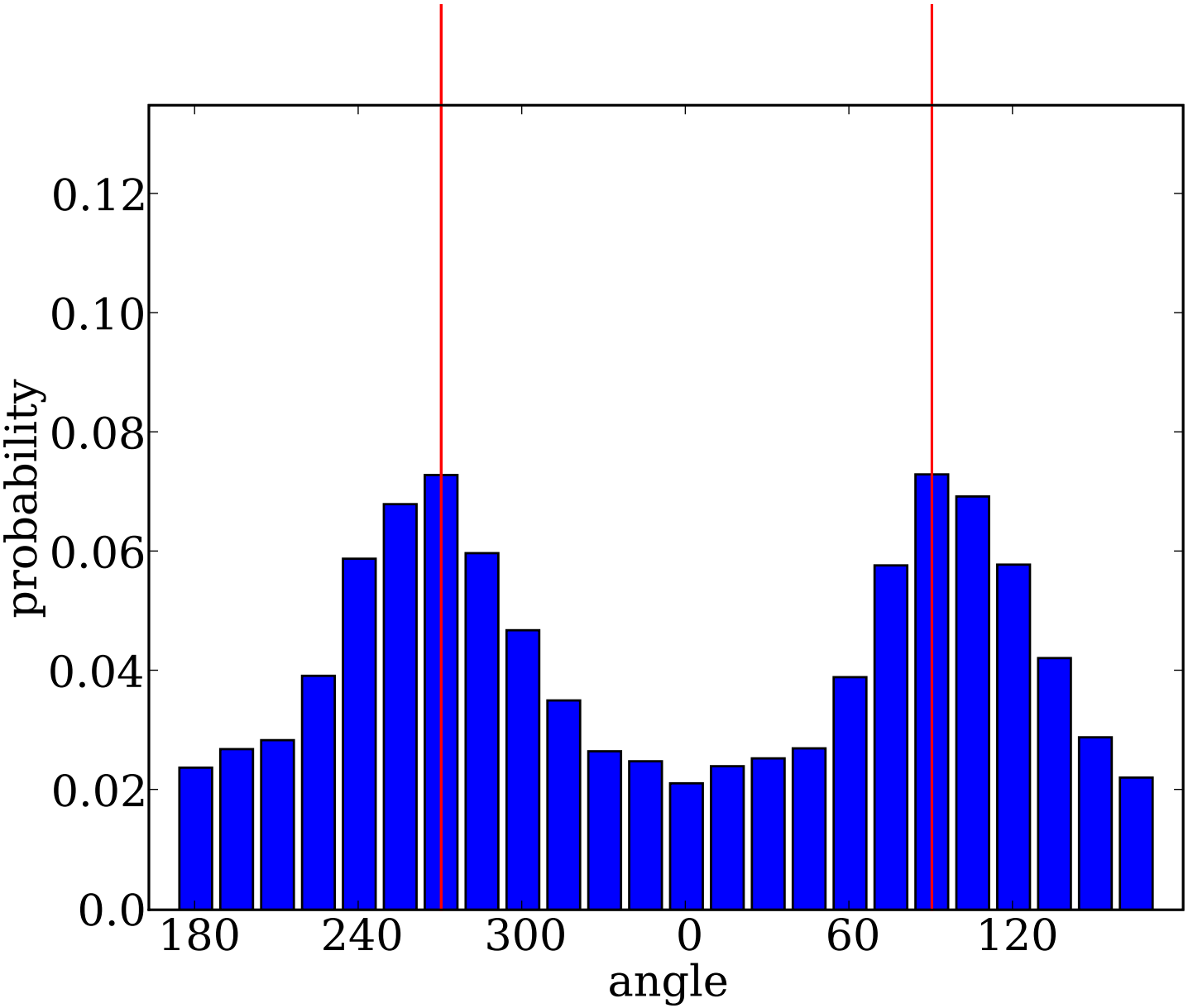
Transparent motion



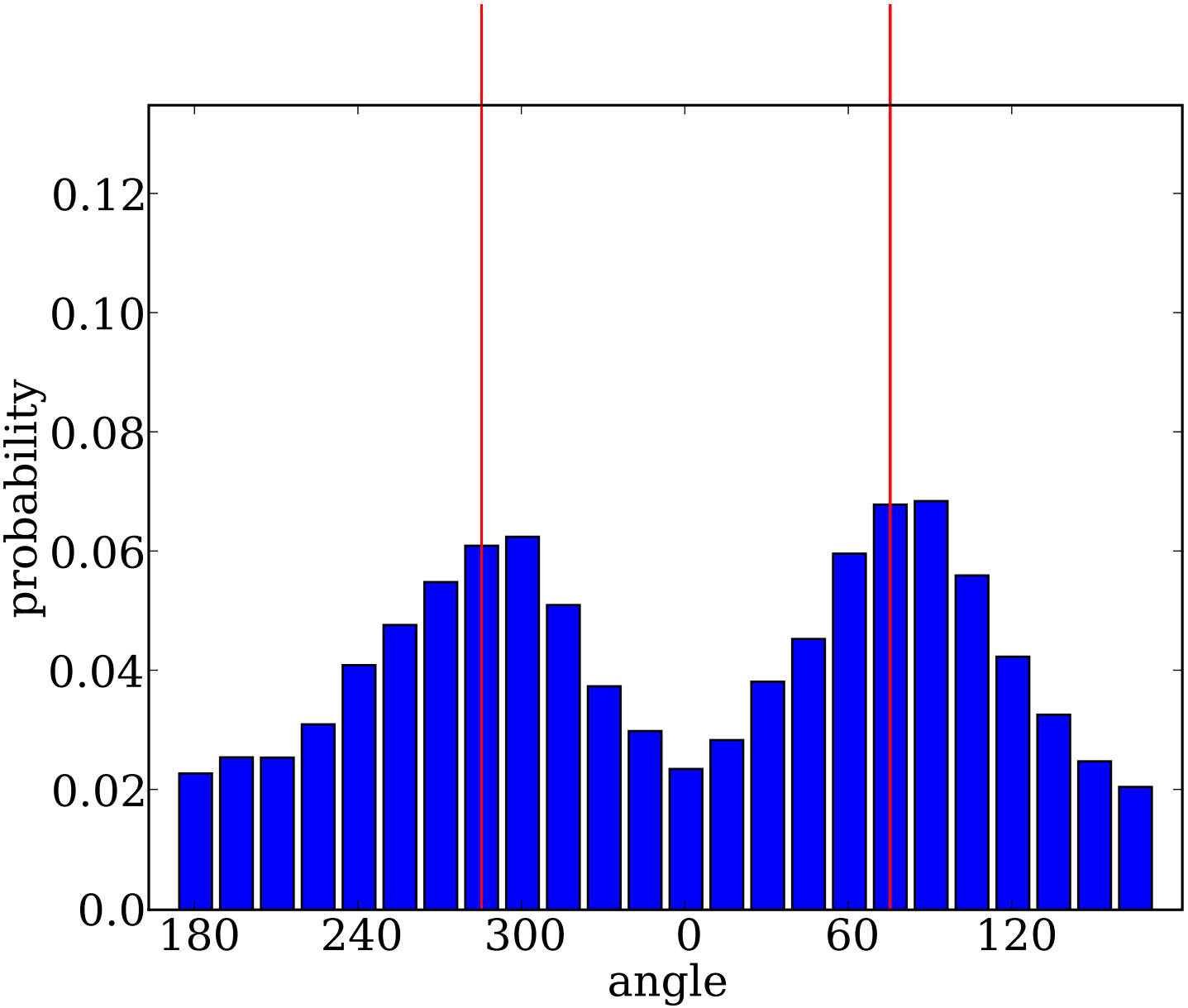
Transparent motion



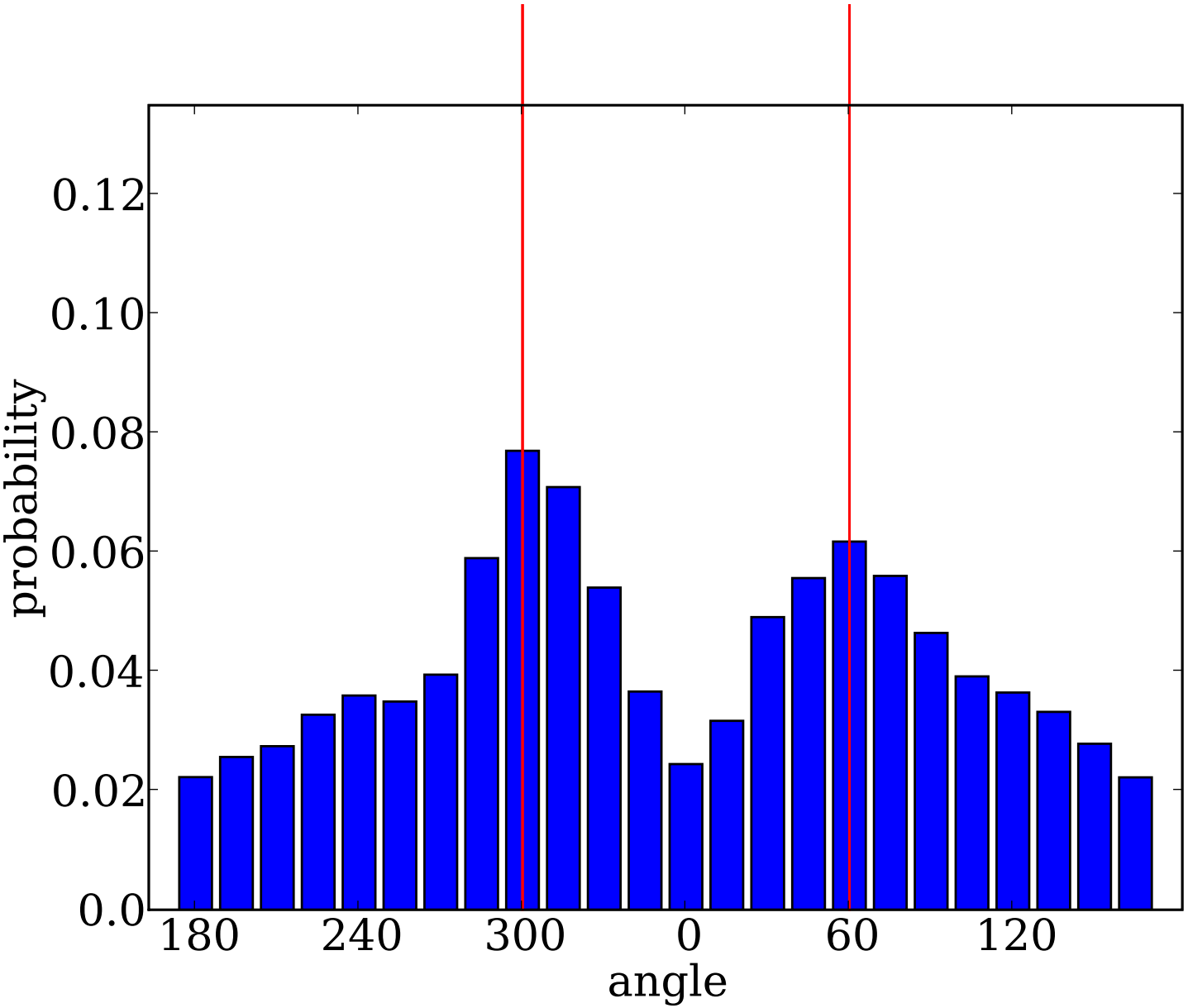
Transparent motion



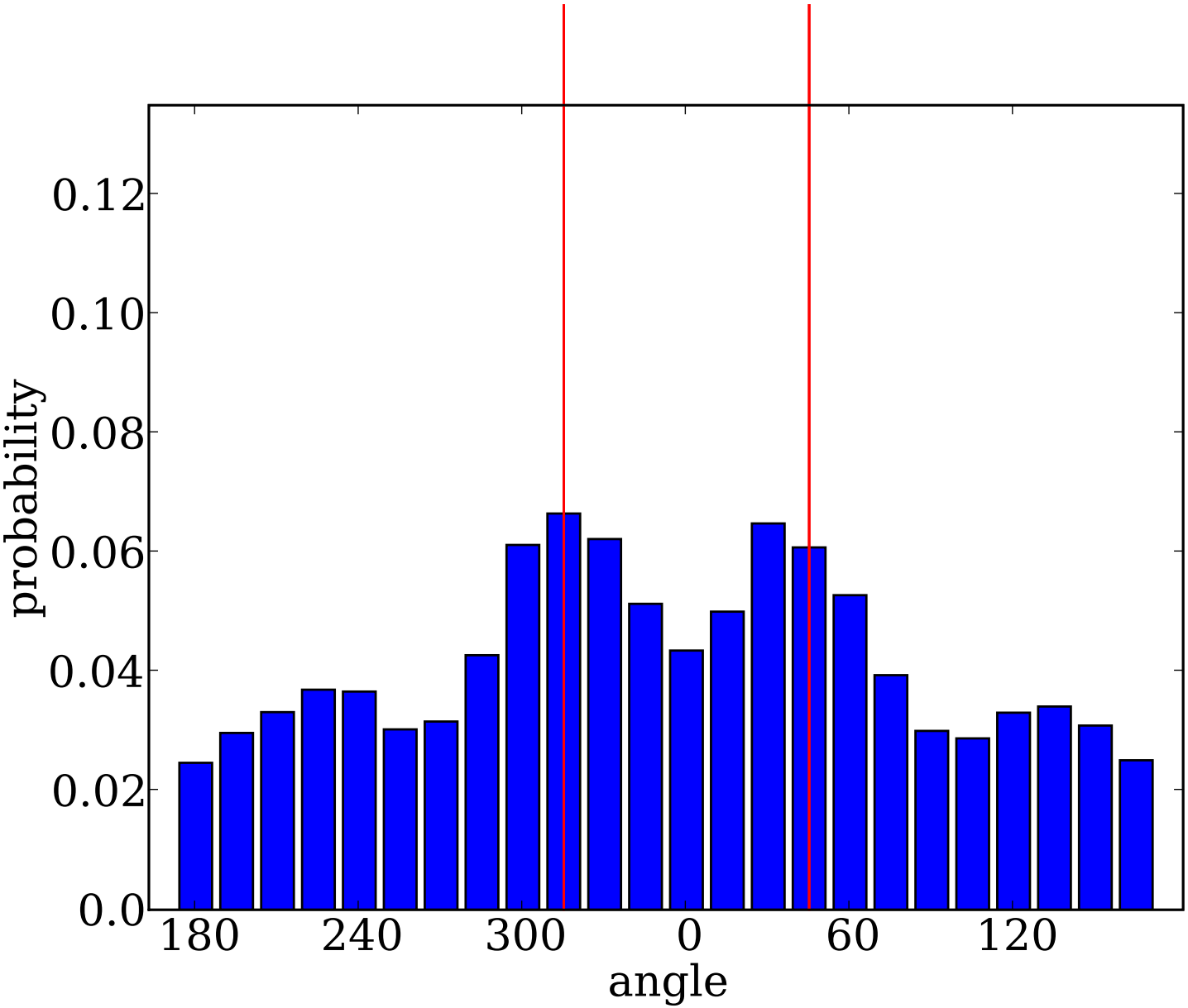
Transparent motion



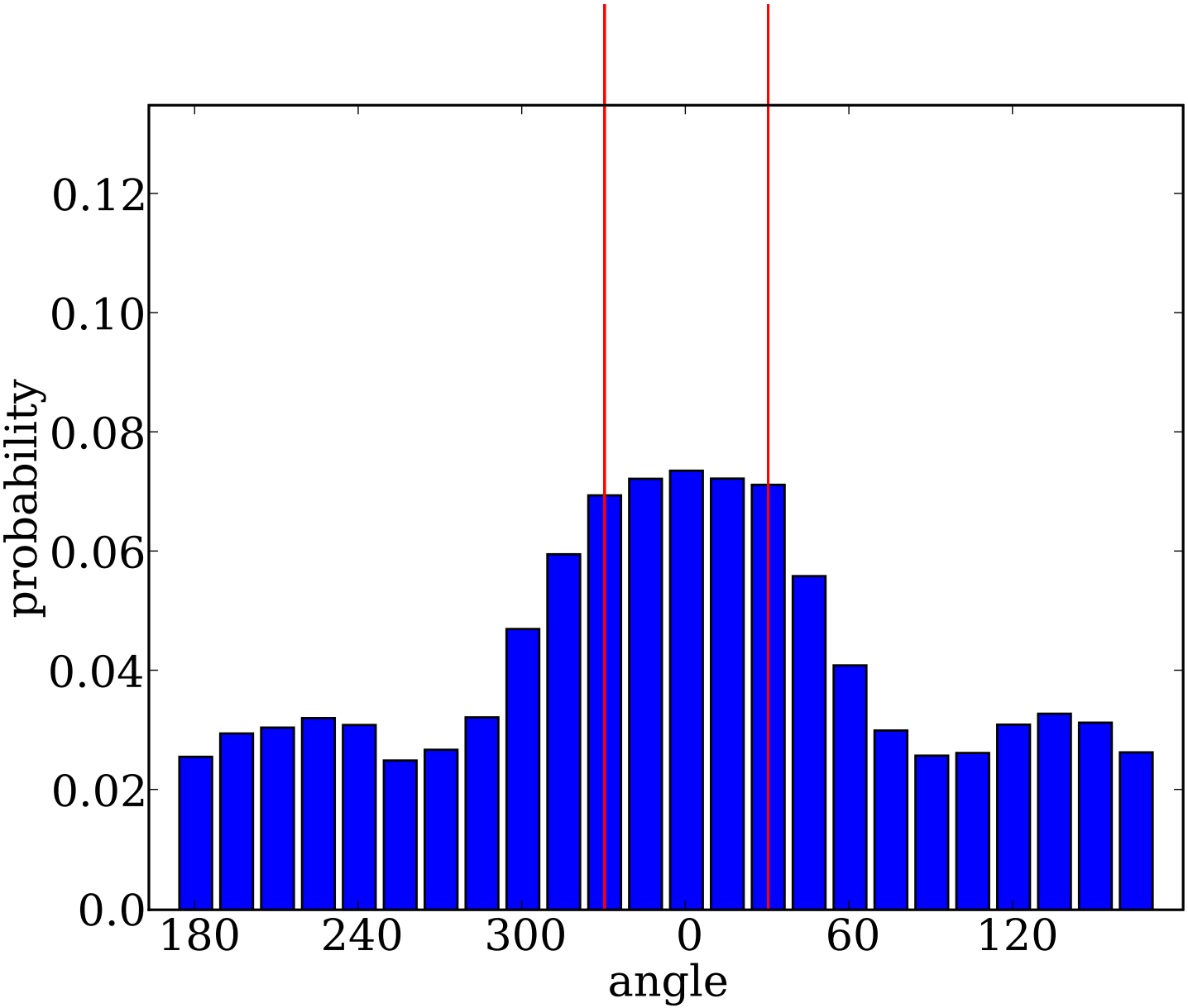
Transparent motion



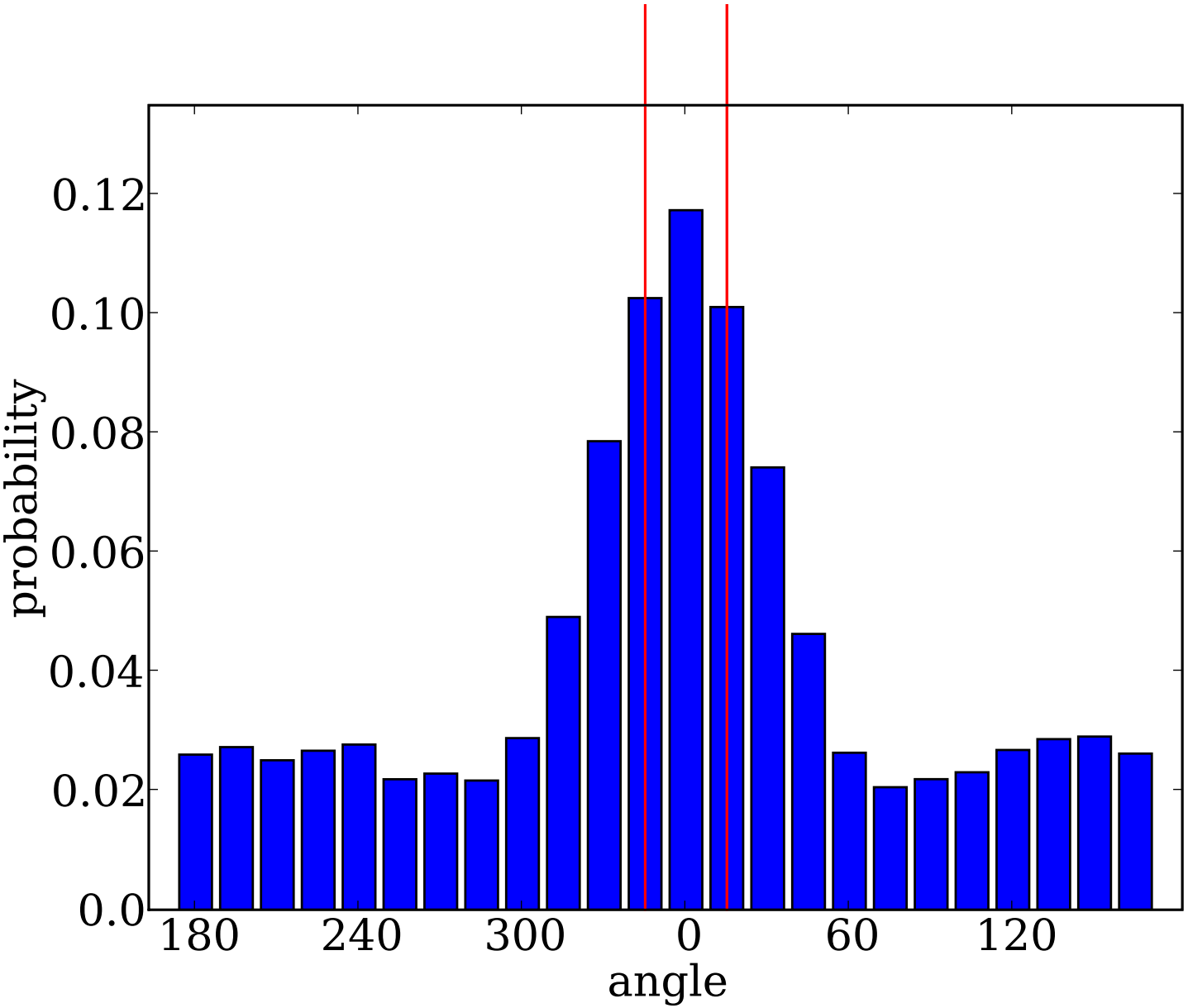
Transparent motion



Transparent motion

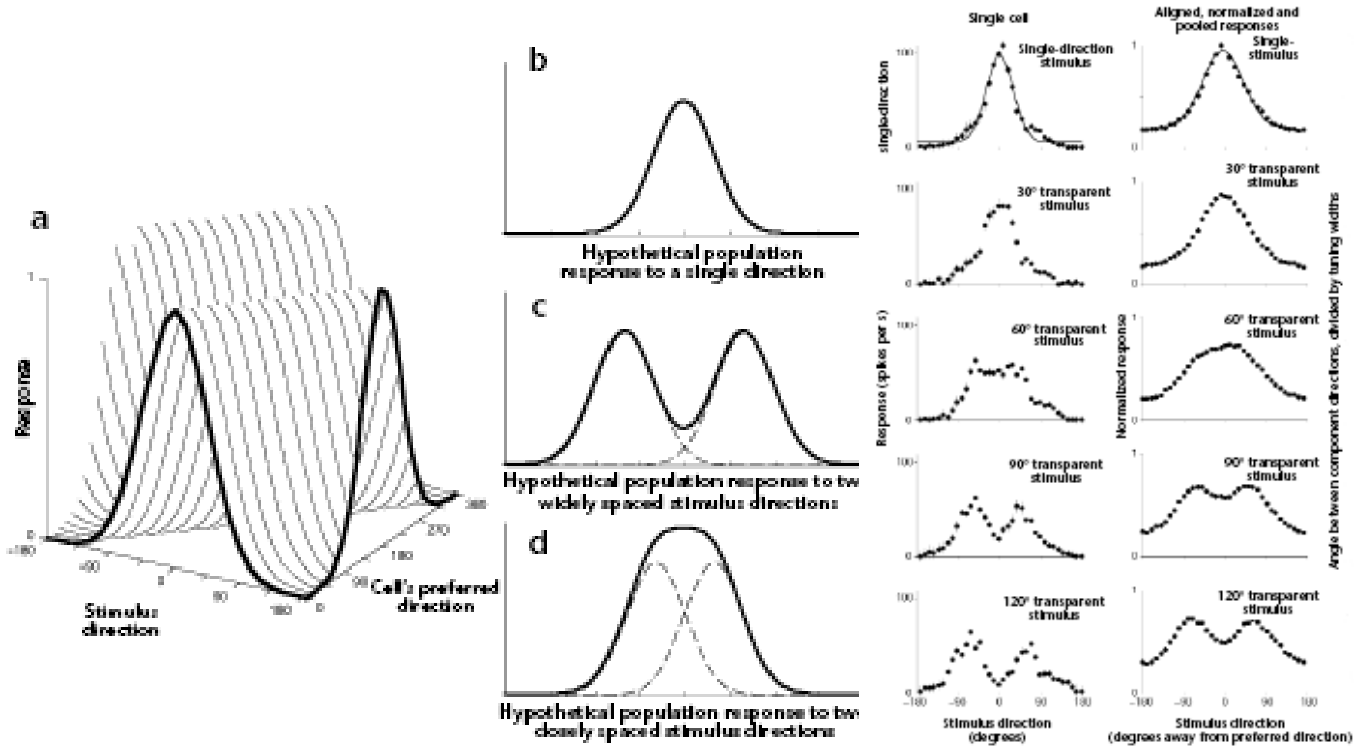


Transparent motion



Some findings shared with biological vision

- Directions easier to discriminate at larger angles.
- Averaging at smaller angles (for example, Treue et al., Nature 2000:)



- Motion repulsion.

Stacking (II): Learning Stereo

- Learning a dictionary for stereo:
- Generate left-right camera pairs with known disparities.
- Top layer: Ground truth.

Stacking (II): Learning Stereo

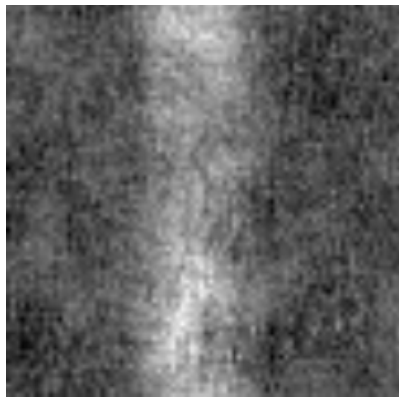
left image



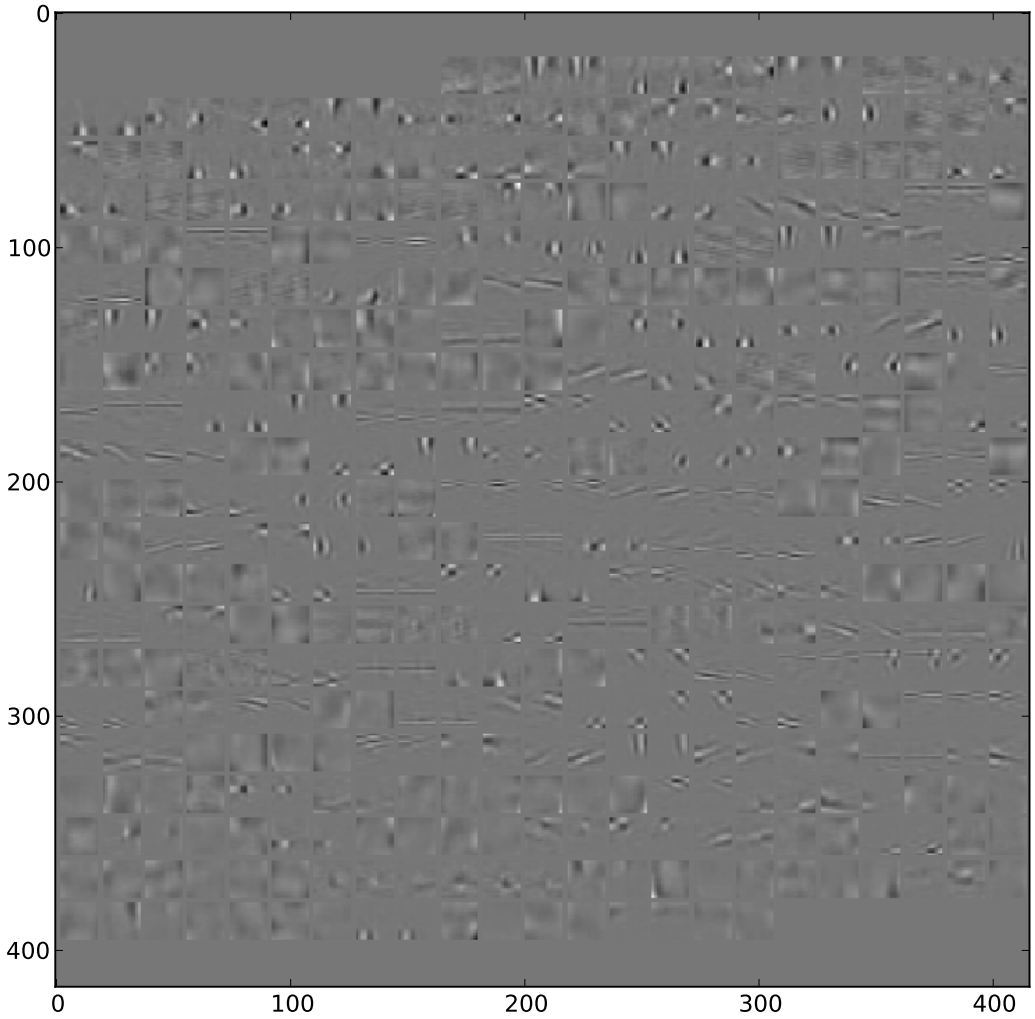
right image



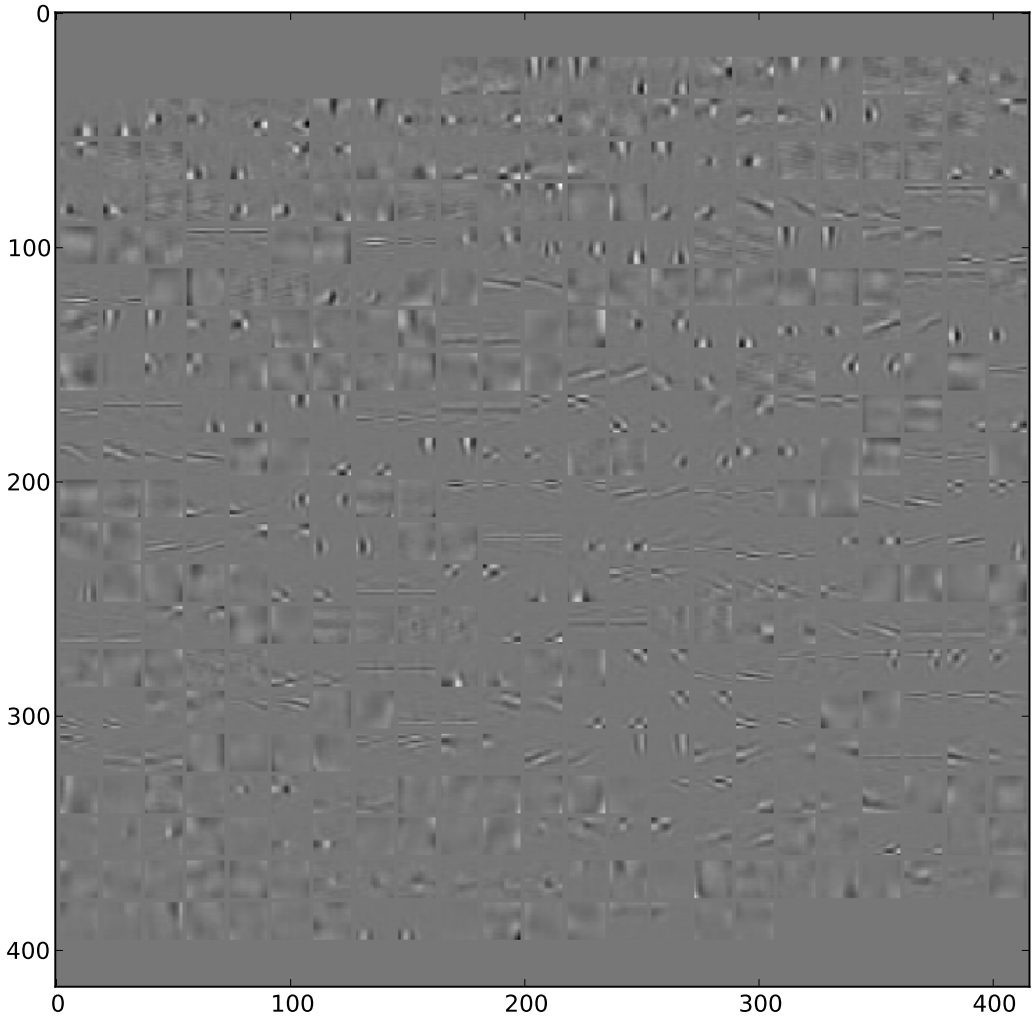
predicted disparity



Learning for stereo matching?



Learning for stereo matching?



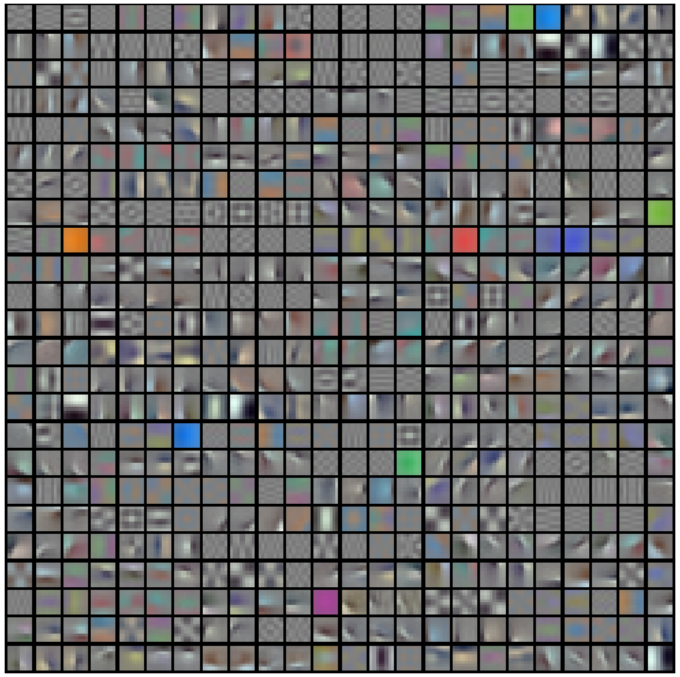
Learning within-image correlation structure

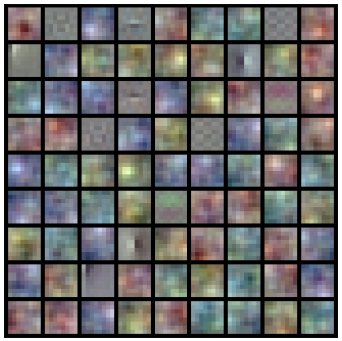
- Learn within-image correlations by setting:

$$x = y$$

- For example, (Karklin et al.), (Ranzato et al.)
- Can "gate" the weights of an auto-encoder to train within-image correlation structure with gradient descent (Memisevic, 2011).

Tiny images





CIFAR object recognition

#factors / #covar-hiddens / #mean-hiddens	Model	Perf (%)
225 / 225 / 0	cRBM	63.6
225 / 225 / 0	cAE	64.5
900 / 225 / 0	cRBM	64.7
900 / 225 / 0	cAE	65.4
576 / 144 / 81	mcRBM	68.2
576 / 144 / 81	mcAE	67.7

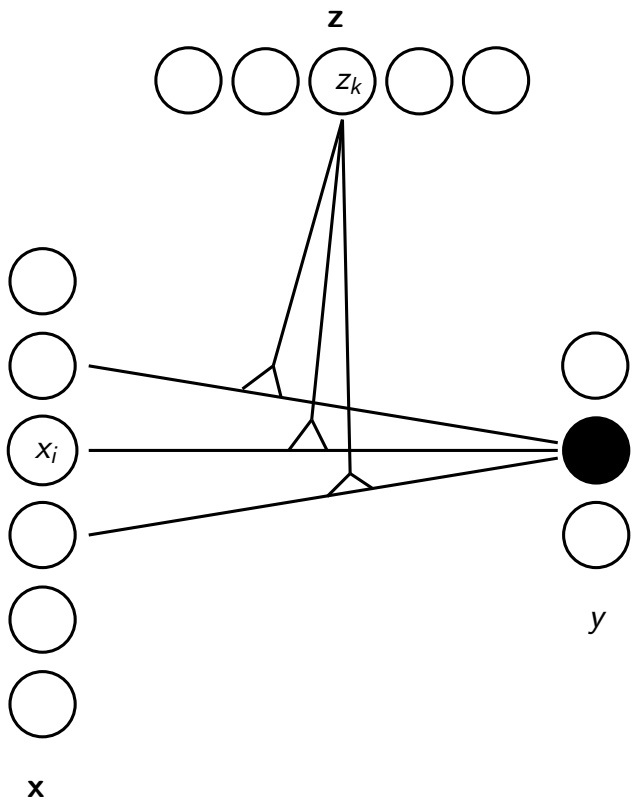
Tabelle: CIFAR-10 recognition rates.

Discussion

- The ability to recognize *relations*, in addition to "things", seems like an immensely useful skill to have.
- "Correspondence" a fairly universal task in vision (and beyond).
- Requires to learn in the presence of "multiplies" (at least in the presence of *squares*).
- This seems like a good reason to have complex cells.
- Interesting future work: more applications; deeper theoretical understanding; faster learning on more data (HPC).

Training 100.000.000.000.000.000 classifiers

A degenerate case: Replace output **image** by a **class-label**



- Model can be shown to be equivalent to a mixture of 2^K logistic regressors with weight sharing.
- Surprisingly, learning and inference ($p(y|\mathbf{x})$) in closed form.

Training 100.000.000.000.000.000 linear classifiers

- This model turns logistic regression:

$$p(y|\mathbf{x}) = \frac{\exp(\mathbf{w}_y^{\mathrm{t}}\mathbf{x})}{\sum_{y'}\exp(\mathbf{w}_{y'}^{\mathrm{t}}\mathbf{x})}$$

- ... into **bi-linear logistic regression**:

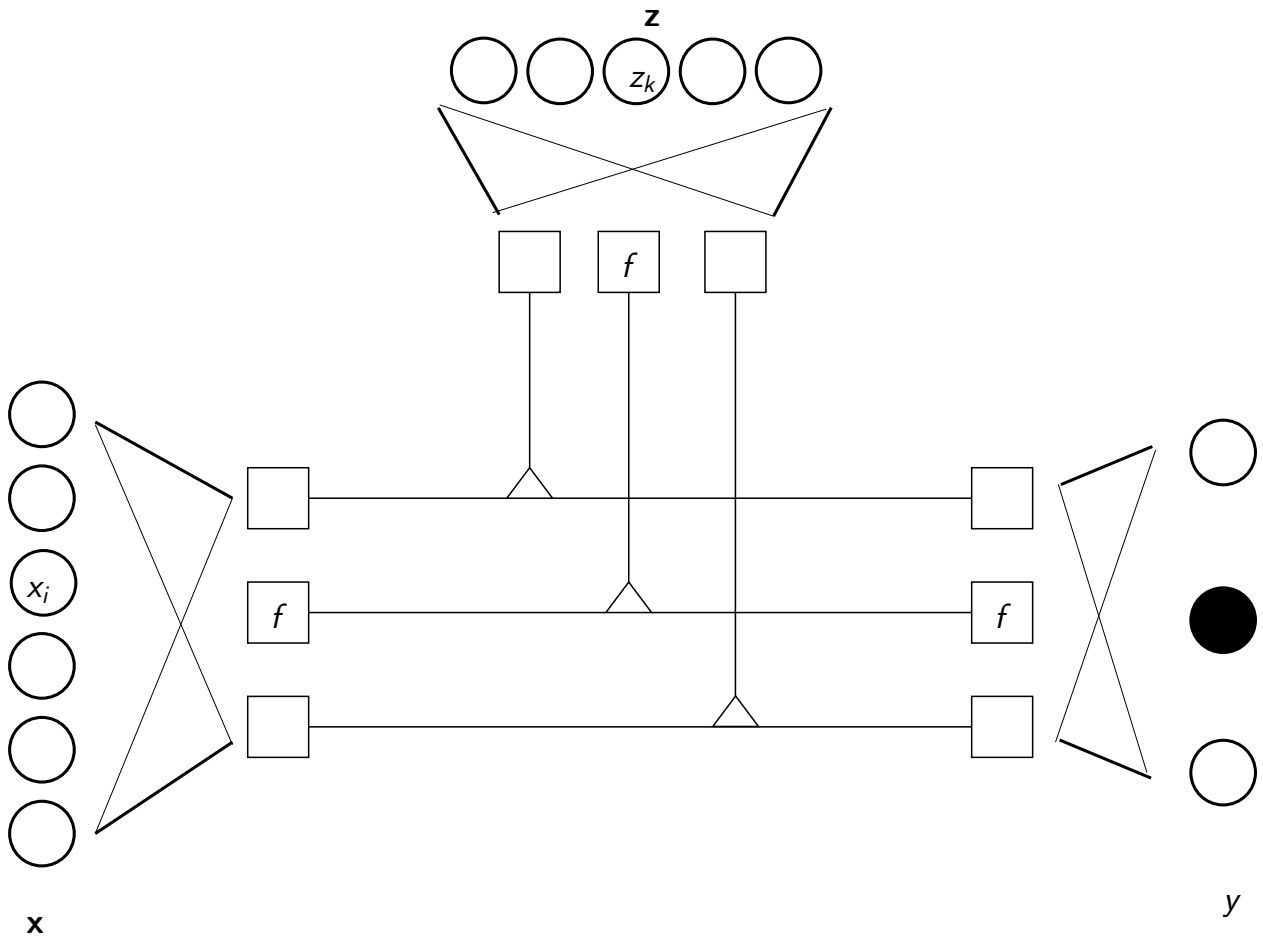
$$p(y,\mathbf{h}|\mathbf{x}) = \frac{\exp(\mathbf{h}^{\mathrm{t}}W_y\mathbf{x})}{\sum_{y'\mathbf{h}'}\exp(\mathbf{h}'^{\mathrm{t}}W_{y'}\mathbf{x})}$$

$$p(y|\mathbf{x}) = \sum_{\mathbf{h}} p(y,\mathbf{h}|\mathbf{x})$$

- Marginalization tractable!

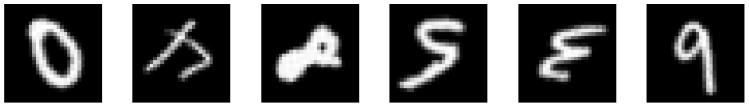
$$\begin{aligned} p(y|\mathbf{x}) &= \sum_{\mathbf{h}} p(y,\mathbf{h}|\mathbf{x}) \propto \sum_{\mathbf{h}} \exp(\mathbf{x}^{\mathrm{t}}W_y\mathbf{h}) \\ &= \sum_{\mathbf{h}} \exp(\sum_{ik} W_{yik}x_i h_k) = \prod_k (1+\exp(\sum_i W_{yik}x_i)) \end{aligned}$$

Factorizing 100.000.000.000.000.000 classifiers

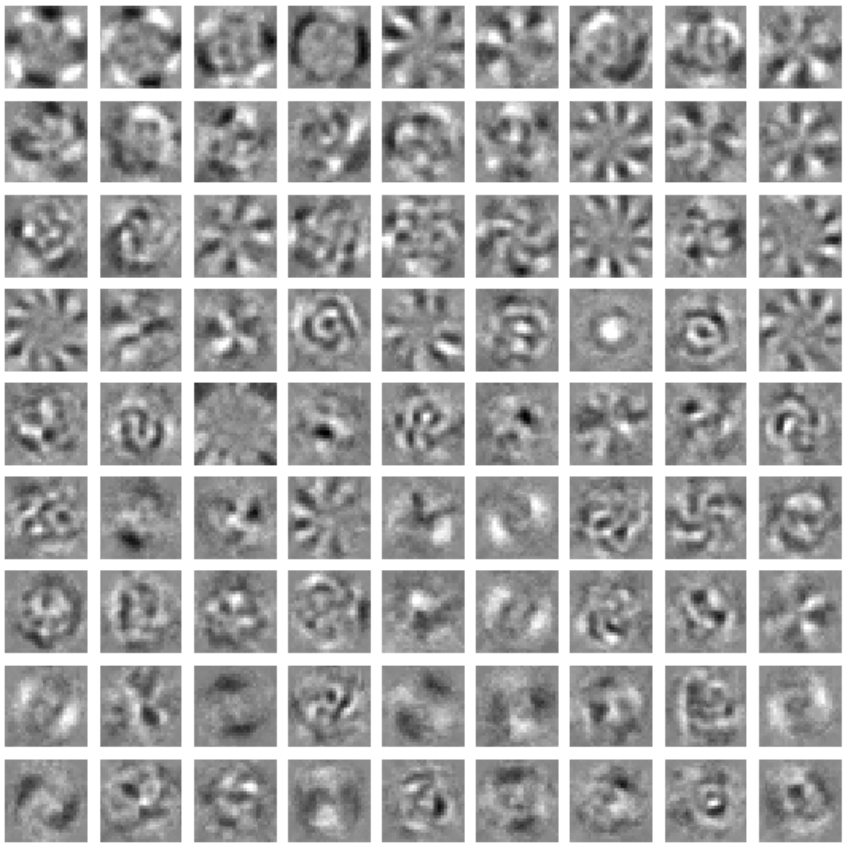


- Can use factorization and other tricks like before.

Rotated digit classification



- Data-set from the "deep learning"-challenge [Larochelle et al., 2007].
- Learned rotation-invariant filters:



Classification error rates

	SVMs		NNet	RBM	DEEP		GSM	
dataset/model:	SVMRBF	SVMPOL	NNet	DBN1	DBN3	SAA3	GSM	(unfact)
rectangles	2.15	2.15	7.16	4.71	2.60	2.41	0.83	(0.56)
rect.-images	24.04	24.05	33.20	23.69	22.50	24.05	22.51	(23.17)
mnistplain	3.03	3.69	4.69	3.94	3.11	3.46	3.70	(3.976)
convexshapes	19.13	19.82	32.25	19.92	18.63	18.41	17.08	(21.03)
mnistbackrand	14.58	16.62	20.04	09.80	06.73	11.28	10.48	(11.89)
mnistbackimg	22.61	24.01	27.41	16.15	16.31	23.00	23.65	(22.07)
mnistrotbackimg	55.18	56.41	62.16	52.21	47.39	51.93	55.82	(55.16)
mnistrot	11.11	15.42	18.11	14.69	10.30	10.30	11.75	(16.15)

- Model fully probabilistic.
- Training online, on very large datasets.
- Let 100.000.000.000.000.000 logistic regressors compete with one ∞-dimensional SVM.
- Model is like optimal Bayesian RBM classifier.